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Autonomous and Affiliated to Bharathiar University
Accredited with A++ grade by NAAC,
Peelamedu, Coimbatore – 641004

AUTOMATED APPROVAL WORKFLOW SYSTEM USING SALESFORCE

Internship work submitted to PSGR Krishnammal College for women in partial fulfillment of the requirements for the award of the degree of

Master of Science in Data Analytics

Bharathiar University, Coimbatore-641046

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JULY 2026

CERTIFICATE

This is to certify that this internship work entitled **"AUTOMATED APPROVAL WORKFLOW SYSTEM"** submitted to PSGR Krishnammal College for Women, Coimbatore, in partial fulfillment of the requirements for the award of the Degree of Master of Science in Data Analytics, Bharathiar University, Coimbatore, is a record of original work carried out by **SUBHIKSHA N (25SMDA049)** during her period of study in the Department of Data Analytics (PG) at PSGR Krishnammal College for Women, Coimbatore, under my supervision and guidance.This internship work has not previously formed the basis for the award of any Degree, Diploma, Associate ship, Fellowship, or any other similar title of any University or Institution.

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Forwarded by

Dr. G. ANITHA, M.C.A., M.Phil., Ph.D.	Dr. S. SASIKALA., M.Sc., M.Phil., Ph.D
Faculty Guide	Head of the Department

DECLARATION

I hereby declare that this internship work entitled **"AUTOMATED APPROVAL WORKFLOW SYSTEM"** submitted to PSGR Krishnammal College for Women, Coimbatore for the award of the Degree of Master of Science in Data Analytics, is a record of original work done by **SUBHIKSHA N (25SMDA049)** under the supervision and guidance of **Dr. G. ANITHA, MCA., M.Phil., Ph.D., Assistant Professor,** Department of Data Analytics (PG) at PSGR Krishnammal College for Women, Coimbatore and that this internship work has not formed the basis for the award of any other Degree / Diploma / Associate ship / Fellowship or any similar title to any candidate of any University.

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Faculty Guide

INTERNSHIP COMPLETION

CERTIFICATE

ELGi Digital Innovation Dojo

EXPERIMENT - START SMALL - SCALE FAST

This is to certify that

SUBHIKSHA N

(Roll No 25SMDA049), a student of PSGR Krishnammal College for women, Coimbatore. Pursuing Second year MSc. DA has undergone internship at **ELGi's Digital Innovation Dojo**.

From **23 Feb 2026** to **27 May 2026.**

We wish her success in her future endeavors.

Date : 30 May, 2026
Place : Coimbatore

Mahantesh Kamalapur
DGM – Digital Innovation Dojo

ELGI
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ELGI INNOVATION DOJO

SUBHIKSHA N - 25SMDA049

WORK DIARY

Date	Work Done
23/02/2026	Induction Program – Company introduction and internship orientation.
24/02/2026 26/02/2026	Trailhead Learning – Salesforce CRM Basics and Cloud Computing Fundamentals.
27/02/2026 – 04/03/2026	Agentforce Learning – AI concepts, Salesforce Agentforce features, and hands-on practice.
05/03/2026 – 10/03/2026	Salesforce Administration – Standard Objects, Custom Objects, Fields, and Object Manager.
11/03/2026 16/03/2026	Data Modeling – Lookup Relationships, Master-Detail Relationships, Validation Rules, and Page Layouts.
17/03/2026 – 21/03/2026	Lightning App Development – Creating the Event Management application and configuring navigation.
24/03/2026 – 28/03/2026	Custom Object Development – Event, Registration, Quotation, and Payment objects.
31/03/2026 – 04/04/2026	Record Management – Creating sample records and testing object relationships.
07/04/2026 – 11/04/2026	Salesforce Flow Builder – Learning Flow Builder and creating record-triggered flows.
14/04/2026 – 18/04/2026	Email Automation – Creating email templates and automating email notifications.
21/04/2026 – 25/04/2026	Approval Process – Creating quotation approval workflows and testing approvals.
28/04/2026 – 02/05/2026	Reports and Dashboards – Creating reports and dashboards for Event Management CRM.
05/05/2026 – 09/05/2026	Testing – Unit Testing, Integration Testing, and fixing identified issues.

K Sathya Kumar
6/8/26

12/05/2026 – 16/05/2026	UI Enhancement – Improving page layouts, Lightning pages, and user navigation.
19/05/2026 – 23/05/2026	Documentation – Preparing project report, screenshots, flow diagrams, and testing documents.
26/05/2026 – 27/05/2026	Final Project Demonstration, Project Submission, Mentor Review, and Internship Completion.

K Sathya Kumari
6/8/26

CONTENT

S.NO	TITLE	PAGE NO
	ACKNOWLEDGEMENT	**I**
	ABSTRACT	**II**
1	**INTRODUCTION**	**2**
	1.1 ORGANIZATION PROFILE	**3**
	1.2 PROBLEM STATEMENT	**5**
	1.3 LITERATURE REVIEW	**6**
	1.4 TOOL DESCRIPTION	**8**
	1.5 DOMAIN	**12**
2	**DATA MODELING**	**14**
	2.1 DATA COLLECTION	**15**
	2.2 DATA DESCRIPTION	**15**
	2.3 PROCESS FLOW	**19**
	2.4 DATA PREPROCESSING	**23**
3	**WORKFLOW AUTOMATION**	**26**
	3.1 APPROVAL WORKFLOW	**26**
	3.2 SALESFORCE FLOW AUTOMATION	**27**
	3.3 APPROVAL PROCESS CONFIGURATION	**28**
	3.4 EMAIL NOTIFICATIONS	**33**
4	**IMPLEMENTATION**	**38**
	4.1 APPROVAL AUTOMATION	**38**
	4.2 REJECTION AUTOMATION	**42**
	4.3 ACCOUNTS	**47**
	4.4 REPORTS	**49**
5	**CONCLUSION**	**53**
6	**FUTURE ENHANCEMENTS**	**54**
	REFERENCES	**55**

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"Success is to be measured not so much by the position that one has reached in life but as by the obstacle which he had overcome while trying to succeed".

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ABSTRACT

The Automated Approval Workflow System is a cloud-based application developed using the Salesforce CRM platform to automate and simplify organizational approval processes. Traditional approval methods often depend on manual documentation, emails, and spreadsheets, which can lead to delays, communication gaps, and inefficient tracking of approval requests. This project addresses these challenges by providing an automated workflow that improves efficiency, transparency, and decision-making.

The application is developed using Salesforce Flow, Approval Processes, Custom Objects, Apex, Lightning Web Components (LWC), Reports, and Dashboards. These technologies automate the routing of approval requests, validate user inputs, update application status, and generate automated email notifications. The system enables users to submit loan applications, monitor approval status in real time, and receive instant updates throughout the approval process. Approvers can review, approve, or reject requests through a centralized platform, ensuring consistency and compliance with organizational policies.

The project focuses on automating the complete Loan Approval Workflow, including customer registration, loan application submission, document verification, eligibility assessment, approval routing, manager approval, and customer notification. The system maintains centralized records of all transactions and approval activities, providing better data management, security, and audit tracking.

Overall, the Automated Approval Workflow System demonstrates how Salesforce can be used to build a secure, scalable, and efficient workflow automation solution. By reducing manual effort, minimizing processing time, improving transparency, and enhancing customer experience, the system supports faster and more reliable approval decisions while contributing to the digital transformation of organizational business processes.

1 INTRODUCTION

In today's rapidly evolving digital era, organizations are increasingly adopting automation technologies to improve operational efficiency, reduce manual effort, and enhance decision-making processes. One of the most critical business operations across industries is the approval process, where requests must be reviewed, verified, and authorized before further action can be taken. These approval workflows are commonly used in banking, finance, healthcare, education, manufacturing, and many other sectors. Despite technological advancements, many organizations still rely on manual approval methods involving paper-based documentation, spreadsheets, emails, and physical signatures. Such traditional approaches often lead to processing delays, communication gaps, inconsistent decision-making, and increased administrative workload.

Workflow automation has emerged as an effective solution to overcome these challenges by digitizing and streamlining approval procedures. Automated workflows eliminate repetitive manual tasks, standardize business processes, and ensure that approval requests are routed to the appropriate personnel based on predefined business rules. By reducing human intervention, organizations can improve process accuracy, minimize operational costs, enhance transparency, and provide faster services to customers.

Salesforce is one of the world's leading cloud-based Customer Relationship Management (CRM) platforms, offering powerful tools for workflow automation, application development, business process management, analytics, and customer relationship management. Its low-code and no-code development capabilities enable organizations to automate complex business operations without requiring extensive programming knowledge. Features such as Salesforce Flow, Approval Processes, Validation Rules, Custom Objects, Apex Programming, Lightning Web Components (LWC), and Reports & Dashboards allow developers to build scalable, secure, and highly efficient enterprise applications.

This internship project focuses on the development of an Automated Approval Workflow System using the Salesforce platform. The application is designed to automate the complete loan approval lifecycle by replacing traditional manual approval procedures with an intelligent digital workflow. The system manages every stage of the approval process, including customer registration, loan application submission, document verification, eligibility assessment, approval routing, managerial review, approval or rejection, and customer notification.

The application provides a centralized platform where users can submit loan applications, upload required documents, and monitor the progress of their requests in real time. Based on predefined business rules, Salesforce automatically routes each application to the appropriate loan officer or manager for review. Once an approval decision is made, the system updates the application status automatically and generates email notifications to keep customers and stakeholders informed throughout the approval process. This significantly reduces processing time, improves communication, and enhances overall customer satisfaction.

A major advantage of the Automated Approval Workflow System is its ability to maintain complete transparency and accountability throughout the approval lifecycle. Every action performed within the system is recorded and stored securely in Salesforce, creating a comprehensive audit trail for reporting, compliance, and future reference. Role-based security, validation rules, and standardized workflows ensure that only authorized users can access sensitive information and perform approval-related activities. This improves data integrity while supporting organizational governance and regulatory compliance.

The system also provides analytical reports and interactive dashboards that enable management to monitor workflow performance, analyze approval trends, identify processing bottlenecks, and evaluate employee productivity. These insights help organizations make informed business decisions and continuously improve their operational processes. Since Salesforce is a cloud-based platform, the application is highly scalable and can easily accommodate future business requirements, additional workflow stages, and integration with external systems such as identity verification services, credit score providers, or payment gateways.

The primary objective of this internship is to demonstrate how Salesforce technologies can be utilized to automate business approval workflows efficiently while improving process consistency, operational transparency, and customer experience. The project highlights the practical implementation of workflow automation concepts through Salesforce Flow, Approval Processes, Apex, Lightning Web Components, and reporting tools. It also illustrates how digital transformation can simplify complex approval procedures, reduce manual effort, and support organizations in achieving higher productivity and better service delivery.

The developed Automated Approval Workflow System serves as a reliable, secure, and scalable business solution for modern organizations seeking to automate approval management. By

integrating workflow automation with cloud-based CRM technology, the application enables faster decision-making, improved collaboration, reduced operational costs, and enhanced customer satisfaction. The project demonstrates the significant role of Salesforce in modern enterprise application development and emphasizes the importance of automation in achieving efficient, transparent, and data-driven business operations.

1.1 ORGANIZATION PROFILE

ELGi Equipments Limited is one of the world's leading manufacturers of compressed air systems and air compressor solutions. Established in 1950 and headquartered in Coimbatore, Tamil Nadu, India, the company has built a strong global reputation for delivering innovative, energy-efficient, and reliable compressed air solutions across various industries. Over the years, ELGi has expanded its operations internationally and serves customers in more than 120 countries through an extensive network of distributors and service partners.

The company offers a wide range of products, including rotary screw air compressors, reciprocating air compressors, oil-free air compressors, portable compressors, compressed air accessories, vacuum pumps, and air treatment equipment. These products are widely used in industries such as manufacturing, automotive, textiles, food processing, pharmaceuticals, construction, mining, healthcare, electronics, and engineering. ELGi continuously invests in research and development to deliver advanced technologies that improve operational efficiency, reduce energy consumption, and minimize maintenance costs for customers.

Innovation is one of the core strengths of ELGi Equipments. The organization follows the philosophy of **"Always Better,"** which reflects its commitment to continuous improvement, customer satisfaction, and technological excellence. The company focuses on developing sustainable and intelligent engineering solutions that address modern industrial challenges while maintaining high standards of quality, safety, and environmental responsibility. Through

continuous product innovation and advanced engineering practices, ELGi has established itself as one of the most trusted compressor manufacturers in the global market.

ELGi places significant emphasis on digital transformation and process automation to improve operational efficiency and customer service. The organization adopts modern enterprise technologies such as Customer Relationship Management (CRM) systems, workflow automation, cloud computing, business intelligence, and data analytics to optimize business operations and enhance decision-making. These technologies help streamline internal processes, improve collaboration between departments, and provide customers with faster and more reliable services.

The company also focuses on developing a highly skilled workforce by encouraging continuous learning, technical training, and innovation. Employees are provided with opportunities to work on advanced technologies, participate in research initiatives, and contribute to product development and process improvement. This culture of continuous learning enables ELGi to maintain its competitive advantage in the global market while delivering high-quality products and services.

As part of its digital transformation initiatives, ELGi utilizes automation technologies to simplify business workflows and improve operational transparency. Approval management, customer relationship management, and business process automation play an important role in enhancing productivity and ensuring compliance with organizational policies. These initiatives reduce manual intervention, improve process consistency, and support efficient business operations across different departments.

The internship project titled **"Automated Approval Workflow System"** aligns with ELGi's vision of improving organizational efficiency through automation. The project demonstrates how Salesforce CRM can be used to automate approval workflows, manage customer information, monitor application status, and improve communication among stakeholders. By implementing cloud-based workflow automation, the system supports faster decision-making, enhanced transparency, better resource utilization, and reduced manual effort and processing time.

1.2 PROBLEM STATEMENT

In many organizations, approval processes are still performed using manual methods such as paper-based documentation, email communication, spreadsheets, and physical signatures. These traditional approaches require significant human effort and often lead to delays in processing requests, communication gaps, and inefficient management of approval activities. As the number of customer requests and business transactions increases, manual approval systems become difficult to manage, affecting overall organizational productivity and customer satisfaction.

The loan approval process is one of the most critical business operations in financial institutions. It involves multiple stages, including customer registration, document verification, eligibility assessment, approval by loan officers and managers, and final communication with customers. Managing these activities manually is time-consuming and increases the possibility of errors, duplicate records, misplaced documents, and inconsistent approval decisions. In addition, customers often experience delays because they have limited visibility into the status of their applications.

Another major challenge is the lack of a centralized system for maintaining customer records and approval history. Information stored across multiple documents or spreadsheets makes it difficult to retrieve data, generate reports, and monitor workflow performance. Managers also face difficulties in identifying bottlenecks, tracking pending approvals, and ensuring compliance with organizational policies. Manual notification processes further increase delays, as applicants may not receive timely updates regarding the status of their loan applications.

To address these challenges, organizations require an automated approval management solution that can streamline the entire approval lifecycle while ensuring accuracy, transparency, and security. Workflow automation reduces repetitive manual tasks, standardizes approval procedures, and ensures that applications are routed automatically to the appropriate approvers based on predefined business rules. Real-time status tracking and automated notifications improve communication between customers, loan officers, and management, resulting in faster decision-making and a better customer experience.

The proposed **Automated Approval Workflow System**, developed using the Salesforce platform, provides a centralized and cloud-based solution for managing loan approval processes.

The system automates customer registration, application submission, document verification, approval routing, status updates, and email notifications. It also maintains a secure repository of customer records and approval history, enabling organizations to improve operational efficiency, reduce processing time, ensure compliance, and enhance transparency throughout the approval lifecycle.

Key Problems

- Manual approval processes leading to delays in loan processing.
- Dependence on paper-based documents, emails, and spreadsheets.
- High possibility of human errors and duplicate records.
- Lack of centralized customer and approval data management.
- Limited visibility into application status and approval progress.
- Delayed communication between customers and approvers.
- Difficulty in monitoring workflow performance and pending approvals.
- Inefficient reporting and lack of real-time business insights.
- Challenges in maintaining compliance, security, and audit records.
- Increased operational costs due to repetitive manual activities.

1.3 LITERATURE REVIEW

[1] Dumas, M., La Rosa, M. (2018) discussed the importance of Business Process Management (BPM) in improving organizational performance. Their study explains how workflow automation standardizes business processes, minimizes manual intervention, and enables organizations to execute tasks more efficiently. The authors concluded that automated workflows improve productivity, process transparency, and service quality.

[2] Salesforce Inc. (2023) highlights Salesforce as one of the leading cloud-based Customer Relationship Management (CRM) platforms that supports workflow automation through Salesforce Flow, Approval Processes, Apex, and Lightning Web Components (LWC). The

platform enables organizations to automate approval routing, maintain centralized business records, and provide real-time visibility into business operations while ensuring scalability and security.

[3] Gartner Research (2023) reported that organizations adopting workflow automation experience significant improvements in operational efficiency, faster approval cycles, reduced processing costs, and improved customer satisfaction. The study emphasizes that cloud-based workflow platforms have become an essential component of modern digital transformation strategies.

[3] Microsoft (2023) explained that workflow automation solutions help organizations eliminate repetitive manual activities and improve collaboration between different departments. Their study highlighted that automated approval systems provide better monitoring capabilities, reduce communication delays, and generate real-time business insights through reporting and analytics.

[4] Forrester Research (2022) emphasized that low-code development platforms accelerate application development while reducing implementation complexity. The study concluded that organizations using low-code workflow automation platforms can rapidly build secure and scalable business applications without extensive programming effort, enabling faster digital transformation.

Key Difference

Most of the existing studies focus on workflow automation, business process management, and cloud technologies as general concepts. While these studies explain the benefits of automation and digital transformation, they do not specifically implement an end-to-end Loan Approval Workflow System using Salesforce.

The proposed project extends these concepts by developing a complete Automated Approval Workflow System that automates the entire loan approval lifecycle, including customer registration, loan application submission, document verification, eligibility assessment, approval routing, manager approval, status tracking, and automated email

notifications. The system integrates Salesforce Flow, Approval Processes, Apex, Lightning Web Components (LWC), Reports, and Dashboards into a centralized cloud platform. This implementation not only improves operational efficiency but also enhances transparency, security, compliance, and customer experience through real-time workflow monitoring and automated communication.

1.3 TOOL DESCRIPTION – SALESFORCE CRM

Salesforce is one of the world's leading cloud-based Customer Relationship Management (CRM) platforms, developed to help organizations manage customer relationships, automate business processes, and improve operational efficiency. It provides a secure and scalable environment where businesses can develop custom applications, automate workflows, analyze data, and manage customer interactions without investing in expensive on-premise infrastructure.

Unlike traditional software applications, Salesforce follows the Software as a Service (SaaS) model, allowing users to access the platform through a web browser from anywhere with an internet connection. The platform automatically handles infrastructure management, software updates, security, and data backup, enabling organizations to focus on business operations rather than system maintenance.

Salesforce supports organizations across multiple industries, including banking, healthcare, education, manufacturing, retail, insurance, and government. It offers various cloud services such as Sales Cloud, Service Cloud, Marketing Cloud, Commerce Cloud, Experience Cloud, and Platform Services. These services enable organizations to manage customer information, automate business processes, improve communication, and deliver better customer experiences.

In this internship project, Salesforce serves as the primary development platform for building the Automated Approval Workflow System. The application utilizes Salesforce to manage customer records, process loan applications, automate approval routing, maintain approval history, generate notifications, and monitor workflow performance. By using

Salesforce's declarative development tools and cloud capabilities, the application provides a secure, scalable, and efficient workflow automation solution.

One of the major strengths of Salesforce is its low-code and no-code development environment. Administrators and developers can build applications using drag-and-drop tools such as Flow Builder, Approval Processes, Object Manager, and Lightning App Builder while also extending functionality through Apex programming and Lightning Web Components. This flexibility allows organizations to rapidly develop business applications with minimal coding while still supporting complex business requirements.

Salesforce also provides advanced reporting and dashboard capabilities that help organizations monitor workflow performance, analyze business trends, and generate meaningful insights. Built-in security features such as role-based access control, field-level security, record sharing, validation rules, and audit trails ensure that sensitive business information remains protected while maintaining compliance with organizational policies.

Overall, Salesforce provides a modern cloud platform for developing enterprise applications that improve productivity, reduce operational costs, and support digital transformation initiatives. Its extensive automation capabilities make it an ideal platform for implementing workflow-based applications such as the Automated Approval Workflow System.

FEATURES OF SALESFORCE CRM

Cloud-Based Platform

Salesforce is a cloud-native platform that enables users to access applications securely through the internet without installing software locally. All data is stored securely in the cloud, allowing users to access the system anytime and from any location while ensuring high availability and reliability.

Workflow Automation

Salesforce automates repetitive business activities using declarative tools such as Flow Builder and Approval Processes. Business rules can be configured to automatically assign tasks, update records, send notifications, and route approval requests, reducing manual effort and improving process efficiency.

Approval Process Management

The Approval Process feature enables organizations to automate multi-level approval workflows. Applications can be routed to designated approvers based on predefined conditions, ensuring consistency, transparency, and compliance throughout the approval lifecycle.

Custom Objects

Salesforce allows developers to create custom objects that store organization-specific information. In this project, custom objects are used to manage customer information, loan applications, approval records, and workflow-related data within a centralized database.

Reports and Dashboards

Salesforce provides interactive reporting tools that enable organizations to visualize business performance. Users can monitor approval status, workflow efficiency, pending requests, and organizational performance through dynamic reports and dashboards.

Data Security

Security is one of the strongest features of Salesforce. The platform provides authentication mechanisms, role-based access control, permission sets, validation rules, audit trails, field-level security, and encrypted cloud storage to protect sensitive organizational data.

Scalability

Salesforce is designed to support organizations of all sizes. As business requirements grow, additional users, workflows, integrations, and custom applications can be added without major infrastructure changes.

Real-Time Notifications

The platform supports automatic email alerts and notifications whenever approval status changes. Customers, loan officers, and managers receive immediate updates regarding submitted applications, pending approvals, approvals, or rejections.

Integration Capabilities

Salesforce easily integrates with third-party applications using APIs and web services. Organizations can connect external systems such as KYC verification platforms, payment gateways, SMS gateways, ERP systems, and banking applications to extend business functionality.

Mobile Accessibility

Salesforce applications are accessible through both web browsers and mobile devices. This enables employees and managers to review approval requests, update records, and monitor workflows while working remotely.

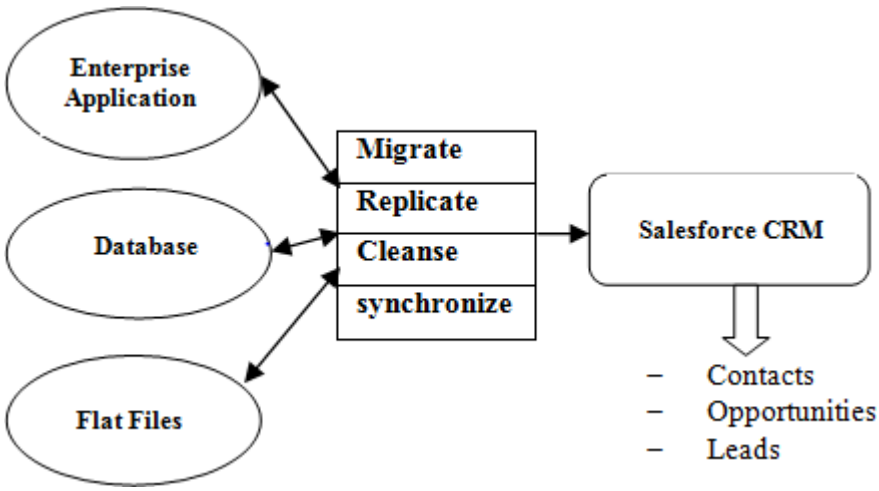


Figure 1.3: Salesforce CRM Platform

Figure 1.3 illustrates the **Salesforce CRM Platform**, which serves as the primary development environment for the Automated Approval Workflow System. The platform provides a comprehensive set of cloud-based tools for managing customer information, automating business workflows, developing custom applications, and generating reports and dashboards. Its low-code capabilities, scalability, and robust security features enable organizations to streamline approval processes, improve operational efficiency, and support digital transformation initiatives.

1.4 DOMAIN – LOAN APPROVAL CRM AND WORKFLOW AUTOMATION

The **Loan Approval CRM and Workflow Automation** domain focuses on improving the efficiency of loan processing by replacing traditional manual approval methods with automated digital workflows. Financial institutions process thousands of loan applications every day, making it essential to have a structured system that can handle customer information, verify documents, evaluate eligibility, and route applications through multiple approval levels. Workflow automation ensures that these activities are performed efficiently while maintaining accuracy, transparency, and compliance with organizational policies.

Traditionally, loan approval involved paper-based forms, manual document verification, physical signatures, spreadsheets, and email communication. These approaches often resulted in long processing times, human errors, duplicate records, lack of visibility, and difficulty in tracking application status. Customers frequently experienced delays in receiving updates regarding their applications, while employees spent considerable time performing repetitive administrative tasks.

With the advancement of cloud computing and enterprise automation platforms such as Salesforce, organizations can now automate approval workflows using predefined business rules. Automated workflows eliminate repetitive manual tasks, assign applications to the appropriate approvers, generate notifications automatically, and maintain a complete history of approval activities. This significantly improves operational efficiency while providing a better customer experience.

The proposed **Automated Approval Workflow System** belongs to the domain of **Business Process Automation (BPA)** and **Loan Approval Management**. The system automates every stage of the loan approval lifecycle, beginning with customer registration and loan application submission, followed by document verification, eligibility assessment, approval routing, managerial review, approval or rejection, and customer notification. Each stage is monitored electronically, ensuring faster processing and better accountability.

Salesforce provides an ideal platform for implementing workflow automation because of its cloud-based architecture, low-code development tools, security mechanisms, and integration capabilities. Using Salesforce Flow, Approval Processes, and Custom Objects, organizations can automate complex approval scenarios while maintaining centralized records and ensuring compliance with business policies.

The Automated Approval Workflow System is designed to provide secure, scalable, and transparent approval management. By reducing manual effort and automating repetitive tasks, the system improves decision-making, minimizes processing time, enhances customer satisfaction, and increases overall organizational productivity.

Application Areas

The **Automated Approval Workflow System** can be applied across various industries that require structured approval processes. Some of the major application areas include:

- Loan application processing in banks and financial institutions.
- Customer onboarding and Know Your Customer (KYC) verification.
- Insurance claim approval workflows.
- Employee leave and reimbursement approval.
- Purchase order and procurement approval.
- Vendor registration and contract approval.
- Educational scholarship and admission approval.
- Healthcare insurance authorization.
- Government service request approvals.
- Enterprise workflow and document management.

2 DATA MODELING

Data modeling is an essential phase in the development of any information system, as it defines how data is organized, stored, and managed within the application. A well-designed data model ensures data consistency, eliminates redundancy, maintains relationships between different entities, and supports efficient business operations. In the **Automated Approval Workflow System**, data modeling plays a significant role in managing customer information, loan applications, approval records, and workflow activities.

The application is developed on the Salesforce platform using a combination of **Standard Objects** and **Custom Objects**. These objects store business data in a structured format while maintaining relationships between different records. The data model enables the application to automate approval workflows, validate user inputs, update application status, and generate reports based on business requirements.

The primary entities involved in this system are **Customer Account** and **Loan Application**. A Customer Account contains the personal and contact information of the applicant, whereas the Loan Application object stores details regarding the requested loan, eligibility status, approval progress, and final decision. These objects are interconnected using lookup relationships, allowing one customer to maintain multiple loan applications while ensuring proper data integrity.

Salesforce automatically maintains data security, validation, audit history, and record ownership, ensuring that only authorized users can access or modify sensitive information. The structured data model also supports workflow automation, approval routing, real-time notifications, reporting, and dashboard generation, making the application efficient, scalable, and reliable.

Overall, the data model serves as the backbone of the **Automated Approval Workflow System** by providing a centralized repository for all business information while supporting automation, transparency, and accurate decision-making throughout the loan approval lifecycle.

2.1 DATA COLLECTION

The **Automated Approval Workflow System** collects and manages information directly through the Salesforce platform. Unlike traditional data analysis projects that rely on external datasets, this application generates and stores operational data in real time as users interact with the system.

The primary source of data is the information entered by customers during the loan application process. Customers provide personal details, contact information, loan requirements, and supporting documents through the application interface. Once submitted, the information is securely stored in Salesforce objects, where it becomes available for verification, approval, reporting, and workflow automation.

Additional information is generated throughout the approval lifecycle. Loan officers update verification details, managers record approval decisions, and the system automatically captures timestamps, approval history, status changes, and notification logs. These records are maintained in a centralized cloud database, ensuring consistency and easy accessibility.

The collected data supports multiple business operations such as customer management, application tracking, approval monitoring, report generation, workflow automation, and performance analysis. Since Salesforce validates records before storage, the system maintains high-quality data while minimizing duplication and human errors.

The centralized approach to data collection improves operational efficiency, enables real-time monitoring, and provides accurate information for business decision-making.

2.2 DATASET DESCRIPTION

The **Automated Approval Workflow System** stores and manages business information using Salesforce **Standard Objects** and **Custom Objects**. Unlike traditional data analytics projects that rely on external datasets, this application captures, stores, and processes data directly within the Salesforce platform. Each object is designed to maintain a specific type of

information, ensuring efficient workflow execution, centralized record management, and secure data storage.

The primary objects used in the application are **Customer Account** and **Loan Application**. These objects are interconnected to support customer registration, loan processing, approval management, and status tracking. Additional workflow information such as approval history, email notifications, and record updates is managed automatically through Salesforce automation tools.

The following objects are used in the application.

1. Customer Account Object

The **Customer Account** is a Salesforce Standard Object used to store the personal and contact details of customers applying for loans. It serves as the primary record for maintaining customer information and acts as the parent object for all loan applications associated with a customer.

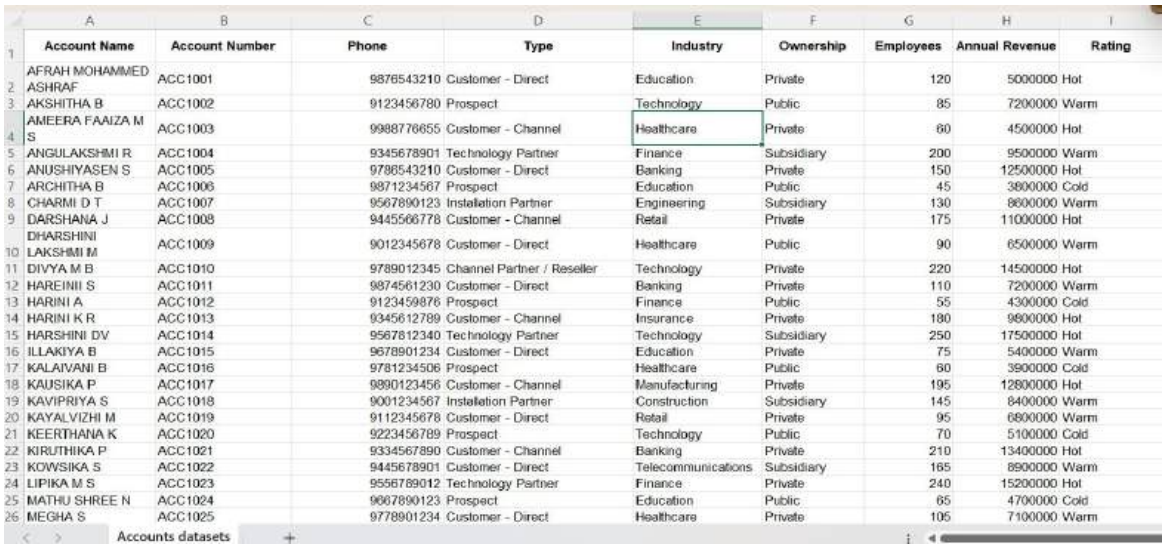
Table 2.2.1: Customer Account Object Fields

Field Name	Description
Account Name	Name of the customer
Account Number	Unique customer account number
Phone	Customer contact number
Email	Customer email address
Industry	Industry category (if applicable)
Ownership	Type of ownership
Annual Revenue	Customer annual income/revenue
Employees	Number of employees
Rating	Customer classification

Table 2.2.1 presents the fields of the Customer Account Object used in the Automated Approval Workflow System. The object stores essential customer information, including personal, contact, and organizational details, which are required for customer registration and loan application processing. It serves as the primary record for maintaining customer data and establishes the relationship between customers and their respective loan applications, ensuring efficient data management and workflow automation within the Salesforce platform**.**

Purpose

- Maintains customer information.
- Supports customer registration.
- Links customer records with loan applications.
- Provides centralized customer management.



Account Name	Account Number	Phone	Type	Industry	Ownership	Employees	Annual Revenue	Rating
AFRAH MOHAMMED ASHRAF	ACC1001	9876543210	Customer - Direct	Education	Private	120	5000000	Hot
AKSHITHA B	ACC1002	9123456780	Prospect	Technology	Public	85	7200000	Warm
AMEERA FAAIZA M S	ACC1003	9988776655	Customer - Channel	Healthcare	Private	60	4500000	Hot
ANGULAKSHMI R	ACC1004	9345678901	Technology Partner	Finance	Subsidiary	200	9500000	Warm
ANUSHIYASEN S	ACC1005	9786543210	Customer - Direct	Banking	Private	150	12500000	Hot
ARCHITHA B	ACC1006	9871234567	Prospect	Education	Public	45	3800000	Cold
CHARMI D T	ACC1007	9567890123	Installation Partner	Engineering	Subsidiary	130	8600000	Warm
DARSHANA J	ACC1008	9445566778	Customer - Channel	Retail	Private	175	11000000	Hot
DHARSHINI LAKSHMI M	ACC1009	9012345678	Customer - Direct	Healthcare	Public	90	6500000	Warm
DIVYA M B	ACC1010	9789012345	Channel Partner / Reseller	Technology	Private	220	14500000	Hot
HAREINII S	ACC1011	9874561230	Customer - Direct	Banking	Private	110	7200000	Warm
HARINI A	ACC1012	9123459876	Prospect	Finance	Public	55	4300000	Cold
HARINI K R	ACC1013	9345612789	Customer - Channel	Insurance	Private	180	9800000	Hot
HARSHINI DV	ACC1014	9567812340	Technology Partner	Technology	Subsidiary	250	17500000	Hot
ILLAKIYA B	ACC1015	9678901234	Customer - Direct	Education	Private	75	5400000	Warm
KALAIVANI B	ACC1016	9781234506	Prospect	Healthcare	Public	60	3900000	Cold
KAUSIKA P	ACC1017	9890123456	Customer - Channel	Manufacturing	Private	195	12800000	Hot
KAVIPRIYA S	ACC1018	9001234567	Installation Partner	Construction	Subsidiary	145	8400000	Warm
KAYALVIZHI M	ACC1019	9112345678	Customer - Direct	Retail	Private	95	6800000	Warm
KEERTHANA K	ACC1020	9223456789	Prospect	Technology	Public	70	5100000	Cold
KIRUTHIKA P	ACC1021	9334567890	Customer - Channel	Banking	Private	210	13400000	Hot
KOWSIKA S	ACC1022	9445678901	Customer - Direct	Telecommunications	Subsidiary	165	8900000	Warm
LIPIKA M S	ACC1023	9556789012	Technology Partner	Finance	Private	240	15200000	Hot
MATHU SHREE N	ACC1024	9667890123	Prospect	Education	Public	65	4700000	Cold
MEGHA S	ACC1025	9778901234	Customer - Direct	Healthcare	Private	105	7100000	Warm

Accounts datasets

Figure 3.2.1: Customer Account Object

Figure 3.2.1 illustrates the Customer Account Object created in Salesforce to store and manage customer information required for the Automated Approval Workflow System. The object contains essential customer details that serve as the primary data source for loan application processing, enabling efficient record management and seamless integration with workflow automation and approval processes.

2. Loan Application Object

The **Loan Application** is a Salesforce **Custom Object** developed specifically for this project. It stores all information related to loan requests submitted by customers and tracks each application throughout the approval process. The object maintains details such as loan amount, application date, approval status, assigned loan officer, and manager approval, enabling efficient workflow automation and centralized loan management.

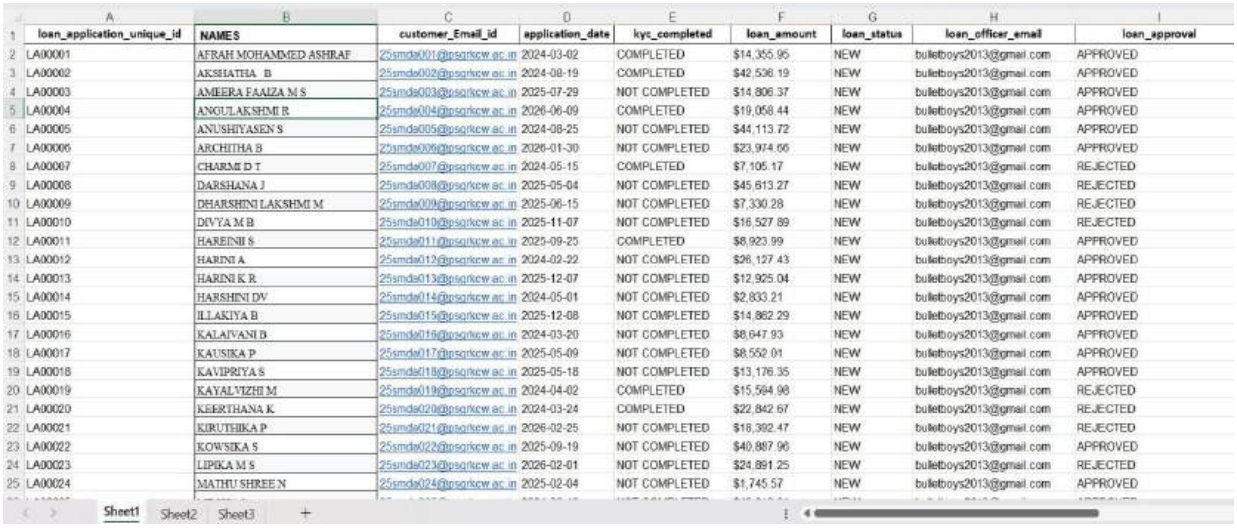
Table 2.2.2: Loan Application Object Fields

Field Name	**Description**
Loan Application Name	Name of the loan application
Loan ID	Unique loan identifier
Application Date	Date of application submission
Customer Email	Applicant's email address
Loan Amount	Requested loan amount
Loan Type	Type of loan
Loan Status	Current approval status
KYC Completed	Verification status
Loan Officer	Assigned loan officer
Manager Approval	Final approval decision

Table 2.2.2 presents the fields of the **Loan Application Object** used in the Automated Approval Workflow System. The object stores all information related to customer loan requests, including application details, loan amount, approval status, verification status, and assigned personnel. It serves as the primary record for monitoring the progress of each loan application and supports workflow automation, approval management, reporting, and decision-making throughout the loan approval lifecycle.

Purpose

- Stores loan application details.
- Tracks the progress of loan approvals.
- Maintains approval history and status updates.
- Supports workflow automation, reporting, and decision-making.



loan_application_unique_id	NAMES	customer_Email_id	application_date	kyc_completed	loan_amount	loan_status	loan_officer_email	loan_approval
LA00001	AFRAH MOHAMMED ASHRAF	25smda001@psgrkcw.ac.in	2024-03-02	COMPLETED	$14,355.95	NEW	bulletboys2013@gmail.com	APPROVED
LA00002	AKSHATHA B	25smda002@psgrkcw.ac.in	2024-08-19	COMPLETED	$42,536.19	NEW	bulletboys2013@gmail.com	APPROVED
LA00003	AMEERA FAAIZA M S	25smda003@psgrkcw.ac.in	2025-07-29	NOT COMPLETED	$14,806.37	NEW	bulletboys2013@gmail.com	APPROVED
LA00004	ANGULAKSHMI R	25smda004@psgrkcw.ac.in	2026-06-09	COMPLETED	$19,058.44	NEW	bulletboys2013@gmail.com	APPROVED
LA00005	ANUSHIYASEN S	25smda005@psgrkcw.ac.in	2024-08-25	NOT COMPLETED	$44,113.72	NEW	bulletboys2013@gmail.com	APPROVED
LA00006	ARCHITHA B	25smda006@psgrkcw.ac.in	2026-01-30	NOT COMPLETED	$23,974.66	NEW	bulletboys2013@gmail.com	APPROVED
LA00007	CHARMI D T	25smda007@psgrkcw.ac.in	2024-05-15	COMPLETED	$7,105.17	NEW	bulletboys2013@gmail.com	REJECTED
LA00008	DARSHANA J	25smda008@psgrkcw.ac.in	2025-05-04	NOT COMPLETED	$45,613.27	NEW	bulletboys2013@gmail.com	REJECTED
LA00009	DHARSHINI LAKSHMI M	25smda009@psgrkcw.ac.in	2025-06-15	NOT COMPLETED	$7,330.28	NEW	bulletboys2013@gmail.com	REJECTED
LA00010	DIVYA M B	25smda010@psgrkcw.ac.in	2025-11-07	NOT COMPLETED	$16,527.89	NEW	bulletboys2013@gmail.com	REJECTED
LA00011	HAREINII S	25smda011@psgrkcw.ac.in	2025-09-25	COMPLETED	$8,923.99	NEW	bulletboys2013@gmail.com	APPROVED
LA00012	HARINI A	25smda012@psgrkcw.ac.in	2024-02-22	NOT COMPLETED	$26,127.43	NEW	bulletboys2013@gmail.com	APPROVED
LA00013	HARINI K R	25smda013@psgrkcw.ac.in	2025-12-07	NOT COMPLETED	$12,925.04	NEW	bulletboys2013@gmail.com	APPROVED
LA00014	HARSHINI DV	25smda014@psgrkcw.ac.in	2024-05-01	NOT COMPLETED	$2,833.21	NEW	bulletboys2013@gmail.com	APPROVED
LA00015	ILLAKIYA B	25smda015@psgrkcw.ac.in	2025-12-08	NOT COMPLETED	$14,862.29	NEW	bulletboys2013@gmail.com	APPROVED
LA00016	KALAIVANI B	25smda016@psgrkcw.ac.in	2024-03-20	NOT COMPLETED	$8,647.93	NEW	bulletboys2013@gmail.com	APPROVED
LA00017	KAUSIKA P	25smda017@psgrkcw.ac.in	2025-05-09	NOT COMPLETED	$8,552.01	NEW	bulletboys2013@gmail.com	APPROVED
LA00018	KAVIPRIYA S	25smda018@psgrkcw.ac.in	2025-05-18	NOT COMPLETED	$13,176.35	NEW	bulletboys2013@gmail.com	APPROVED
LA00019	KAYALVIZHI M	25smda019@psgrkcw.ac.in	2024-04-02	COMPLETED	$15,594.98	NEW	bulletboys2013@gmail.com	REJECTED
LA00020	KEERTHANA K	25smda020@psgrkcw.ac.in	2024-03-24	COMPLETED	$22,842.67	NEW	bulletboys2013@gmail.com	REJECTED
LA00021	KIRUTHIKA P	25smda021@psgrkcw.ac.in	2026-02-25	NOT COMPLETED	$18,392.47	NEW	bulletboys2013@gmail.com	REJECTED
LA00022	KOWSIKA S	25smda022@psgrkcw.ac.in	2025-09-19	NOT COMPLETED	$40,897.96	NEW	bulletboys2013@gmail.com	APPROVED
LA00023	LIPIKA M S	25smda023@psgrkcw.ac.in	2026-02-01	NOT COMPLETED	$24,891.25	NEW	bulletboys2013@gmail.com	REJECTED
LA00024	MATHU SHREE N	25smda024@psgrkcw.ac.in	2025-02-04	NOT COMPLETED	$1,745.57	NEW	bulletboys2013@gmail.com	APPROVED

Figure 3.2.2: Loan Application Custom Object

Figure 3.2.2 illustrates the Loan Application Custom Object developed in Salesforce for the Automated Approval Workflow System. The object is designed to store and manage all information related to customer loan applications, including loan details, application status, verification records, approval decisions, and assigned loan officers. It serves as the central repository for tracking the progress of each loan application throughout the approval lifecycle, enabling efficient workflow automation, real-time status monitoring, and accurate reporting within the Salesforce platform.

2.3 PROCESS FLOW

The **Automated Approval Workflow System** follows a systematic workflow to ensure that every loan application is processed efficiently, accurately, and in accordance with organizational policies. The workflow begins when a customer submits a loan application and continues through

various stages, including document verification, eligibility assessment, approval routing, managerial review, and customer notification. Each stage is automated using Salesforce technologies, reducing manual intervention and enabling faster, more accurate decision-making.

Initially, customer information and loan details are entered into the Salesforce application through user-friendly forms. The system validates the entered information to ensure that all mandatory fields are completed and that the data satisfies predefined business rules. Once the validation process is successfully completed, the application is automatically routed to the assigned loan officer for verification.

The loan officer reviews the submitted information, verifies the supporting documents, and checks whether the applicant meets the eligibility criteria defined by the organization. If any required information is missing or incorrect, the application is returned to the customer for correction. Otherwise, the verified application is forwarded to the manager for the final approval decision.

The manager evaluates the application based on the verification results and organizational policies. After the decision is made, Salesforce automatically updates the application status as **Approved** or **Rejected**. A Record-Triggered Flow then sends an automated email notification to the customer informing them of the final decision. All workflow activities are recorded within Salesforce, providing complete transparency, audit history, and real-time tracking of the approval process.

The process flow adopted in this project improves operational efficiency, minimizes human errors, accelerates approval cycles, and enhances communication between customers and the organization. The automation also enables administrators to monitor application progress, generate reports, and maintain compliance with organizational standards.

Process Flow Diagram

The following process flow diagram illustrates the complete workflow of the Automated Approval Workflow System, beginning with customer registration and ending with the final approval decision and customer notification.

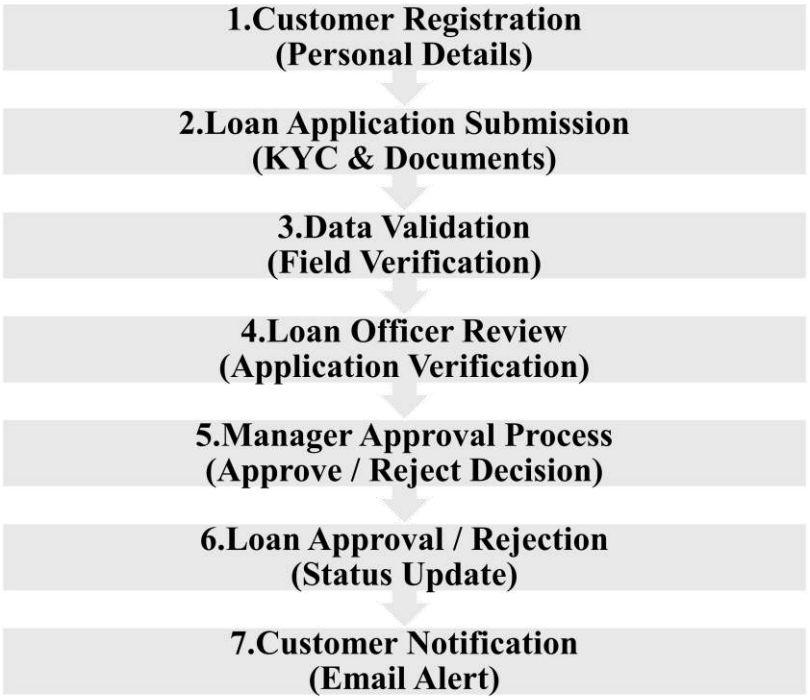


Figure 2.3.1: Workflow of the Automated Approval Workflow System

Figure 2.3.1 illustrates the workflow of the Automated Approval Workflow System. The process automates loan application handling from customer registration to approval decision and customer notification, ensuring an efficient, transparent, and streamlined workflow management process.

Data Flow

The customer account information is first captured and stored in Salesforce. Loan applications are then created and associated with the corresponding customer account. The application data passes through validation, eligibility assessment, approval review, and status update stages. Throughout the workflow, Salesforce Flows and Approval Processes manage record updates, approval routing, and notification generation, ensuring accurate data processing and seamless workflow automation.

Process Steps

Step 1: Customer Registration

The customer registers by entering personal details such as name, email address, phone number, and contact information. The information is securely stored in the Salesforce **Customer Account** object, creating the primary customer record for future loan applications.

Step 2: Loan Application Submission

The registered customer submits a loan application by entering loan details and uploading the required supporting documents. The application is stored in the Salesforce **Loan Application** custom object for further processing.

Step 3: Document Verification

The submitted documents are verified by the loan officer to ensure that all required information is complete, accurate, and valid. This verification process helps maintain data integrity and ensures that only eligible applications proceed to the next stage.

Step 3: Data Validation

Salesforce **Validation Rules** verify mandatory fields, email formats, loan amounts, and other predefined business conditions before allowing the application to proceed further. This prevents incomplete or invalid records from entering the approval workflow.

Step 4: Loan Officer Review

The assigned loan officer reviews the application, verifies customer eligibility, examines the submitted documents, and confirms that all business requirements have been satisfied before forwarding the application to the manager.

Step 5: Manager Approval

The manager evaluates the verified application based on organizational policies and approval criteria. The application is then either **Approved** or **Rejected**, and the decision is recorded automatically within Salesforce.

Step 6: Customer Notification

Based on the manager's decision, Salesforce automatically updates the application status and sends an email notification to the customer informing them whether the loan application has been

approved or rejected. This automated notification ensures timely communication and improves the overall customer experience.

2.3 DATA PREPROCESSING

Data preprocessing is an important stage in any information system, as it ensures that the data stored in the application is accurate, complete, consistent, and ready for further processing. In traditional data analytics projects, preprocessing involves cleaning datasets by removing duplicate records, handling missing values, and transforming data into a suitable format. However, in the **Automated Approval Workflow System**, preprocessing is performed dynamically within the Salesforce platform through validation rules, field constraints, object relationships, and workflow automation before the data is permanently stored.

When a customer submits a loan application, the system validates all mandatory information such as customer details, contact information, loan amount, loan type, and supporting documents. Salesforce **Validation Rules** ensure that incomplete or incorrect records cannot be saved until all required fields are properly completed. This helps maintain data quality and prevents errors during the approval process.

The application also performs format validation for important fields such as email addresses and phone numbers. Invalid entries are rejected immediately, allowing users to correct their information before submission. Business rules are also applied to verify loan amounts, ensuring that only valid values are accepted according to organizational policies.

Duplicate records are minimized through Salesforce's standard duplicate management features and unique record identification. This prevents multiple customer records or duplicate loan applications from being created, thereby improving data consistency and reducing administrative effort.

Lookup relationships between the **Customer Account** and **Loan Application** objects ensure that every loan application is associated with the correct customer. This relationship maintains referential integrity and enables the system to retrieve customer information efficiently throughout the approval lifecycle.

Role-based security and field-level permissions further improve data integrity by restricting unauthorized users from viewing or modifying sensitive information. Only authorized users, such as loan officers, managers, and system administrators, can perform approval-related activities, ensuring confidentiality and compliance with organizational policies.

The system also utilizes Salesforce automation to update application status, maintain approval history, and generate notification records automatically. These automated processes reduce manual intervention, improve workflow consistency, and ensure that data remains synchronized across different stages of the loan approval process.

Overall, the preprocessing mechanisms implemented in Salesforce improve the quality, reliability, and consistency of business data. By validating user inputs, preventing duplicate records, enforcing business rules, and maintaining secure object relationships, the **Automated Approval Workflow System** provides a strong foundation for efficient workflow automation and accurate decision-making.

Table 2.3.1: Data Preprocessing Activities

Activity	**Purpose**
Mandatory Field Validation	Ensures that all required information is entered before submission.
Email Format Validation	Accepts only valid email addresses.
Phone Number Validation	Verifies that phone numbers follow the required format.
Loan Amount Validation	Ensures that the loan amount satisfies predefined business rules.
Duplicate Record Prevention	Prevents duplicate customer and loan application records.
Lookup Relationships	Links loan applications with the correct customer account.
Role-Based Security	Restricts unauthorized access to sensitive information.
Automated Status Updates	Keeps loan application records synchronized throughout the approval process.

Table 2.3.1 presents the data preprocessing activities implemented in the Automated Approval Workflow System. These preprocessing mechanisms ensure data accuracy, maintain consistency, prevent duplicate records, validate user inputs, and enforce business rules before information is stored in Salesforce. As a result, the system supports secure, reliable, and efficient workflow automation throughout the loan approval lifecycle.

3 WORKFLOW AUTOMATION

The **Automated Approval Workflow System** was implemented using the Salesforce platform to automate the complete loan approval process. The implementation involved configuring Salesforce objects, approval workflows, Record-Triggered Flows, email notifications, and application settings to reduce manual intervention and improve operational efficiency.

The system automates customer registration, loan application processing, approval routing, and notification services while maintaining centralized customer records and approval history. Salesforce declarative tools such as **Approval Process**, **Flow Builder**, **Email Templates**, and **Email Alerts** were utilized to implement the workflow without extensive coding.

By integrating these automation features, the application ensures that every loan application follows a standardized approval process. The automated workflow minimizes processing delays, reduces human errors, improves communication between stakeholders, and provides complete visibility into every stage of the loan approval lifecycle.

The implementation also enhances data consistency and operational transparency by automatically updating application records, maintaining approval history, and generating notifications whenever the application status changes. Overall, workflow automation enables organizations to improve productivity, maintain compliance with business policies, and deliver faster and more reliable services to customers.

3.1 APPROVAL WORKFLOW

The **Automated Approval Workflow** follows a structured process that automates the complete loan approval lifecycle. After a customer submits a loan application, the system validates the entered information and routes the application to the assigned loan officer for verification. Once verified, the application is forwarded to the manager for approval. Based on the manager's decision, the application status is updated automatically, and the customer receives an email notification regarding the approval outcome.

This automated workflow minimizes manual processing, improves transparency, and ensures that every loan application follows a standardized approval procedure. The workflow also

maintains a complete approval history, enabling administrators and managers to monitor application progress and ensure compliance with organizational policies.

3.2 SALESFORCE FLOW AUTOMATION

A **Record-Triggered Flow** was implemented in Salesforce to automate customer notifications based on changes to the loan application status. Whenever the loan status is updated, the flow automatically evaluates the record and performs the appropriate action without requiring manual intervention. This automation ensures that every status change is processed consistently and efficiently.

If the application status is **Approved**, the system automatically sends a loan approval email to the customer and updates the application status. If the application status is **Rejected**, the system sends a rejection email and updates the application record accordingly. These automated actions eliminate repetitive manual tasks, reduce processing time, and ensure timely communication with customers.

The Record-Triggered Flow also improves workflow consistency by executing predefined business logic whenever a loan application is created or updated. Since the automation is performed within the Salesforce platform, all workflow activities are recorded and maintained for auditing and reporting purposes. This enables organizations to monitor workflow performance while maintaining transparency and data integrity throughout the loan approval lifecycle.

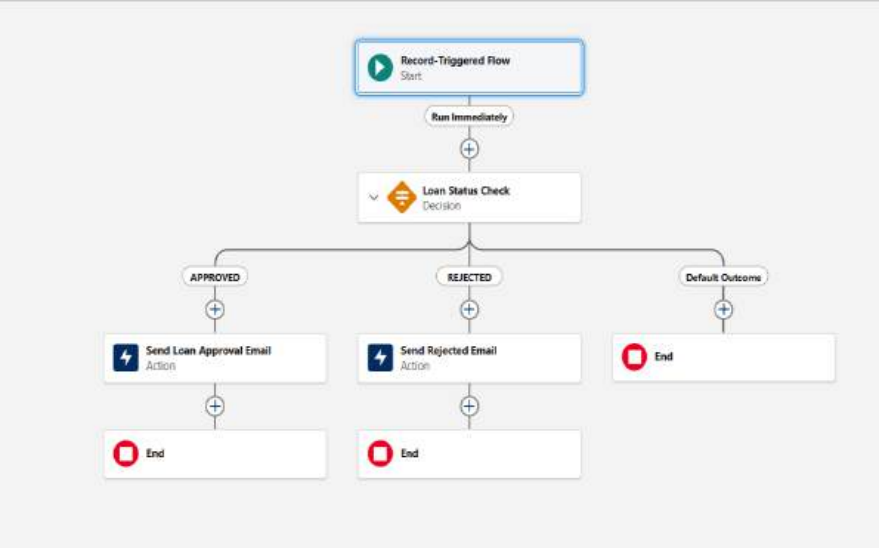


Figure 3.2.1: Salesforce Record-Triggered Flow

Figure 3.2.1 illustrates the Salesforce Record-Triggered Flow used in the Automated Approval Workflow System. The flow is automatically initiated whenever a loan application record is created or updated. Based on the application status, it evaluates predefined business conditions, updates records, sends approval or rejection email notifications, and completes the workflow without requiring manual intervention. This automation improves efficiency, accuracy, consistency, and customer communication throughout the loan approval process.

Workflow Logic

- **Approved**
    - Send Approval Email.
    - Update Loan Status.
    - End Process.
- **Rejected**
    - Send Rejection Email.
    - Update Loan Status.
    - End Process.
- **Other Status**
    - No Action.
    - End Process.

3.3 APPROVAL PROCESS CONFIGURATION

The **Approval Process Configuration** was implemented in Salesforce to automate the routing of loan applications through different stages of the approval lifecycle. Salesforce configuration tools were used to create the application environment, customize objects, design page layouts, and establish approval rules that support efficient workflow automation. These configurations ensure that every loan application follows a standardized approval process while maintaining data accuracy, transparency, and security.

The configuration process includes creating the Salesforce application, customizing objects, designing page layouts, configuring approval processes, and managing workflow settings. These components work together to provide a centralized platform for customer management, loan processing, approval tracking, and notification services.

3.3.1 Salesforce Application Configuration

The Salesforce application was created using **App Manager** to provide a centralized workspace for customer management, loan applications, approval workflows, reports, and notifications. Navigation items, tabs, and user profiles were configured to improve accessibility and provide users with quick access to frequently used features.

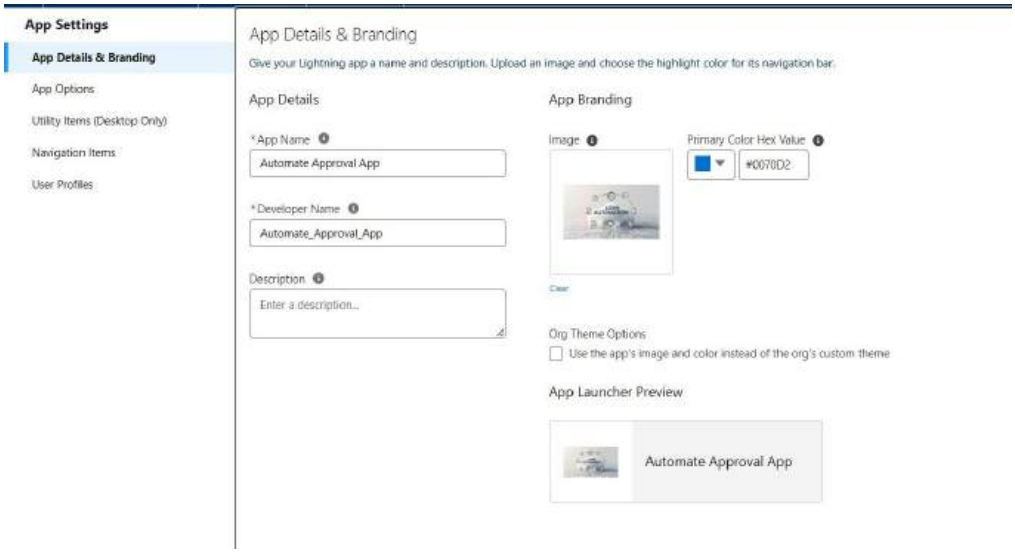


Figure 3.3.1: Salesforce Application Configuration

Figure 3.3.1 illustrates the Salesforce Application Configuration created using App Manager. The application provides a centralized interface for managing customer accounts, loan applications, approval workflows, reports, and notifications. Proper configuration of navigation menus and user access improves usability and enables efficient management of the Automated Approval Workflow System.

3.3.2 Object Management

The **Object Manager** was used to create the **Loan Application** custom object and configure the **Customer Account** standard object. Custom fields, lookup relationships, page layouts, validation rules, and object settings were implemented to support the approval workflow and ensure efficient data management.

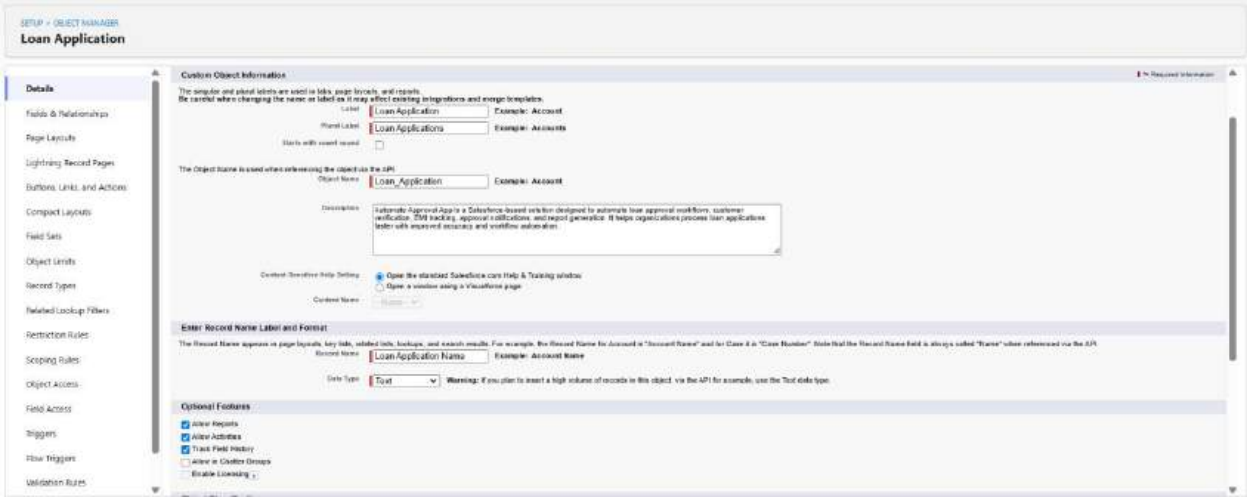


Figure 3.3.2: Salesforce Object Manager

Figure 3.3.2 illustrates the Salesforce Object Manager, which is used to create, configure, and manage standard and custom objects within the Salesforce platform. It provides options to define fields, relationships, page layouts, validation rules, and automation settings, enabling efficient customization of the data model to support the Automated Approval Workflow System.

3.3.3 Customer Account Configuration

The **Customer Account** object stores customer information such as name, contact number, email address, account details, and other essential information required during the loan approval process. It serves as the parent object for all loan applications and enables centralized customer record management.

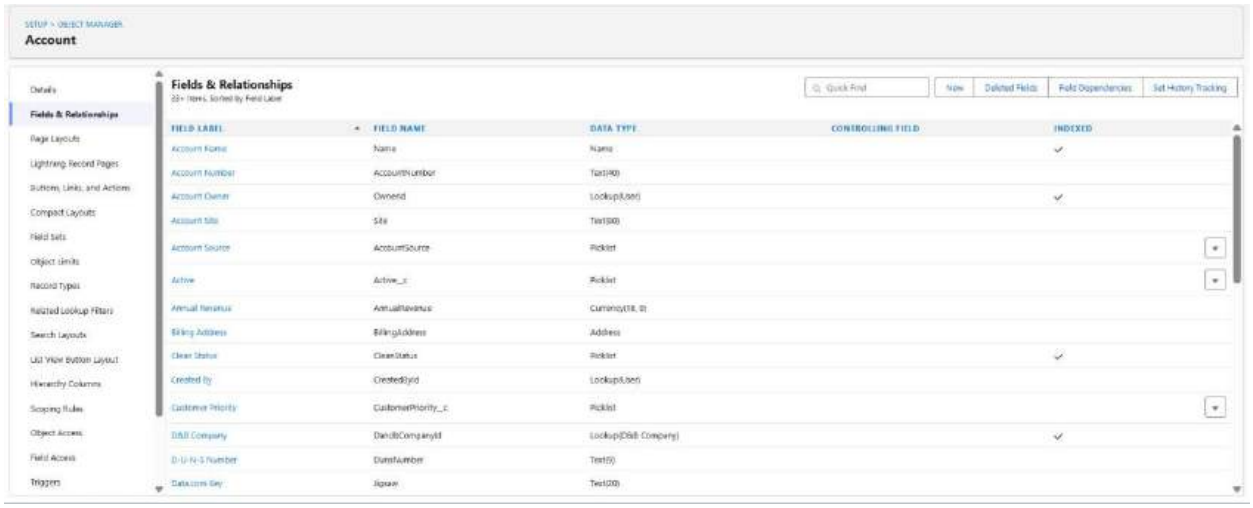


Figure 3.3.3: Customer Account Configuration

Figure 3.3.3 illustrates the configuration of the Salesforce Customer Account object used in the Automated Approval Workflow System. The configuration includes customer information fields, page layouts, and object settings that enable efficient customer registration, record management, and integration with loan applications.

3.3.3 Loan Application Page Layout

The **Loan Application** page layout was customized to organize loan details, customer information, approval status, KYC status, assigned loan officer information, and related records in a structured manner. The customized layout improves usability, simplifies navigation, and enables users to process loan applications efficiently.

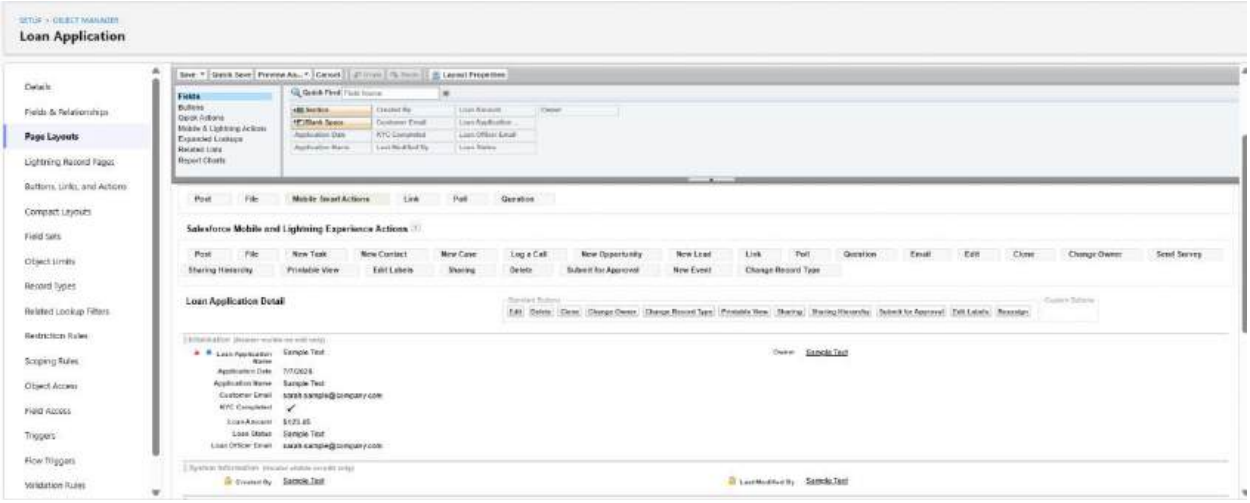


Figure 3.3.3: Loan Application Page Layout

Figure 3.3.3 illustrates the customized Loan Application page layout developed in Salesforce. The layout organizes customer information, loan details, approval status, verification details, and related records into a user-friendly interface, improving data accessibility and supporting efficient loan processing.

3.3.4 Approval Process

The **Approval Process** automatically routes loan applications to the designated approver based on predefined business rules. After the approver reviews the application, Salesforce updates the application status, records the approval history, and initiates the corresponding workflow actions such as email notifications and status updates.

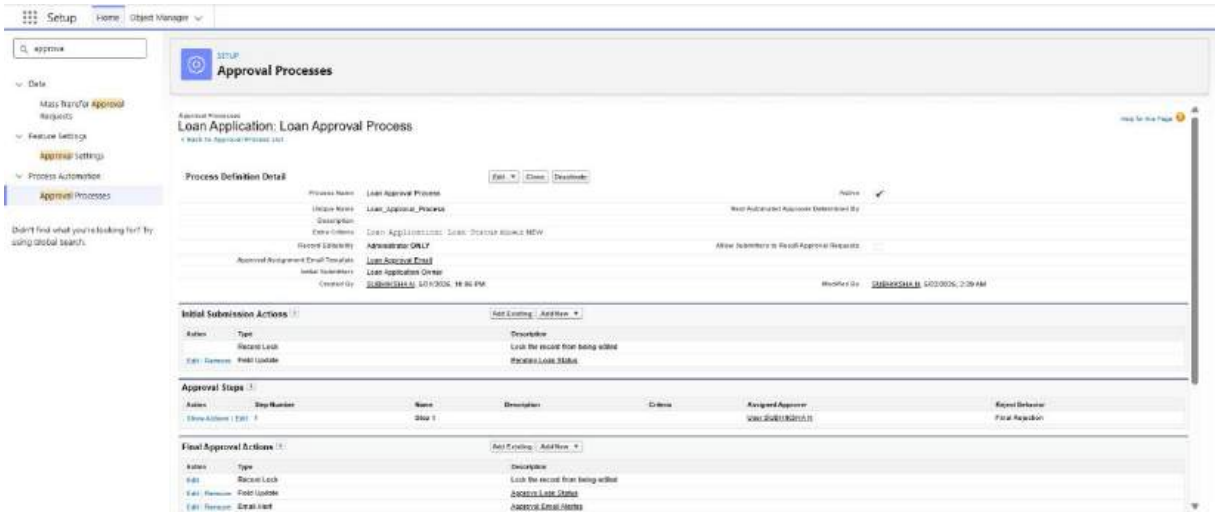


Figure 3.3.4: Approval Process Configuration

Figure 3.3.4 illustrates the Approval Process configuration implemented in Salesforce for the Automated Approval Workflow System. The configuration automates the routing of loan applications to the appropriate approvers, records approval decisions, updates application status, and maintains approval history, ensuring a secure, transparent, and standardized approval workflow.

3.3 EMAIL NOTIFICATIONS

Email notifications were automated using Salesforce **Email Templates** and **Email Alerts** to ensure that customers receive timely updates regarding the status of their loan applications. The notification system automatically sends emails whenever there is a change in the application status, eliminating the need for manual communication and improving customer engagement.

The email notification process is integrated with the Salesforce Approval Process and Record-Triggered Flow. Based on the manager's approval decision, the system identifies the appropriate email template and sends it automatically to the customer's registered email address. This automation ensures that applicants receive immediate updates regarding the progress and final outcome of their loan applications.

Automated email notifications improve communication, reduce processing delays, enhance transparency, and provide customers with a better user experience. They also minimize

administrative effort while ensuring that all notification activities are recorded within the Salesforce platform for future reference and auditing purposes.

3.3.1 Approval Email Template

The **Approval Email Template** is automatically sent when a loan application is approved. The template informs the customer that the loan request has been successfully approved and provides confirmation regarding the approval decision. The standardized template ensures consistent communication and delivers timely updates to customers.

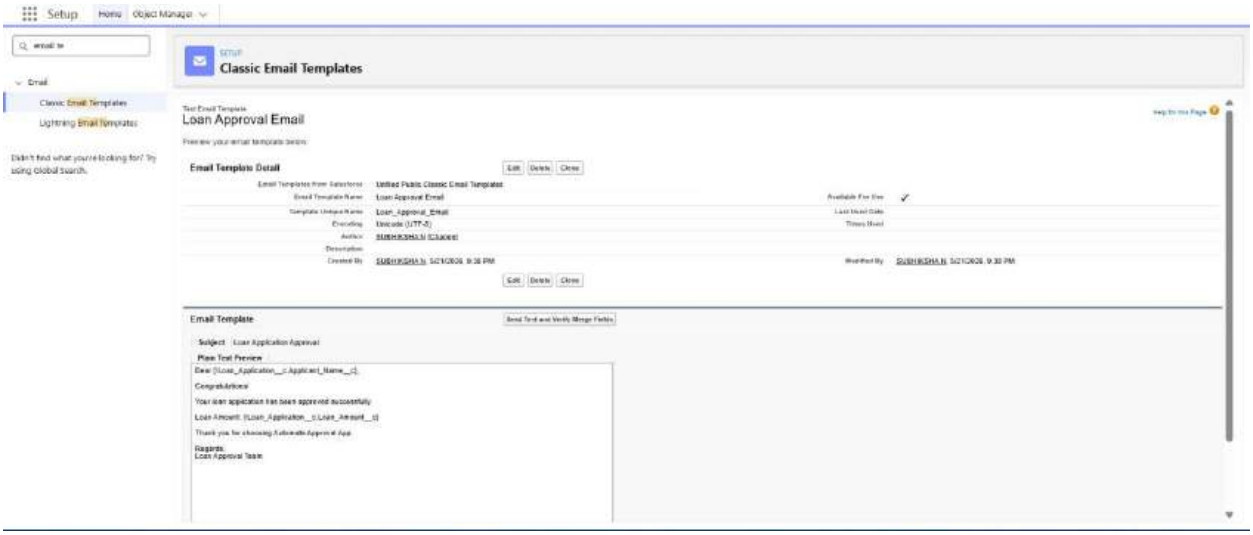


Figure 3.3.1: Approval Email Template

Figure 3.3.1 illustrates the Approval Email Template configured in Salesforce. The template is automatically triggered after a loan application is approved and provides customers with a formal confirmation of the approval decision. It ensures clear, consistent, and professional communication throughout the approval process.

3.3.2 Rejection Email Template

The **Rejection Email Template** is automatically sent when a loan application is rejected. It informs the customer that the loan request could not be approved and communicates the final

decision in a professional manner. Automated rejection emails ensure prompt communication and maintain transparency throughout the workflow.

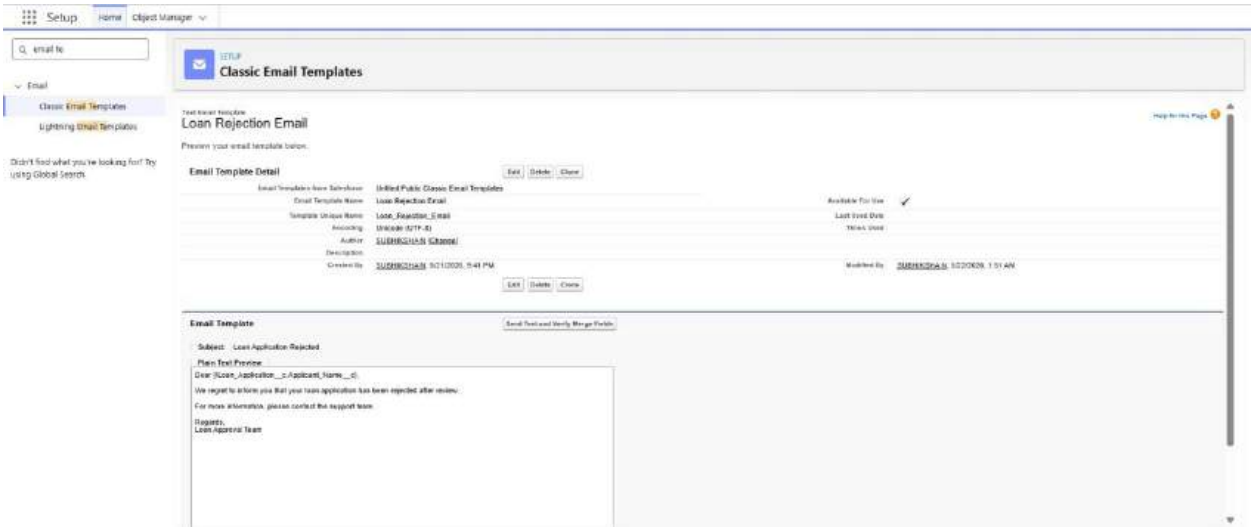


Figure 3.3.2: Rejection Email Template

Figure 3.3.2 illustrates the Rejection Email Template configured in Salesforce. The template is automatically sent whenever a loan application is rejected, ensuring that customers are informed promptly about the final decision while maintaining consistent and professional communication.

3.3.3 Approval Email Alert

The **Approval Email Alert** is configured to automatically send the approval email whenever the application status changes to **Approved**. The alert is integrated with the Record-Triggered Flow and Approval Process, enabling immediate notification without manual intervention.

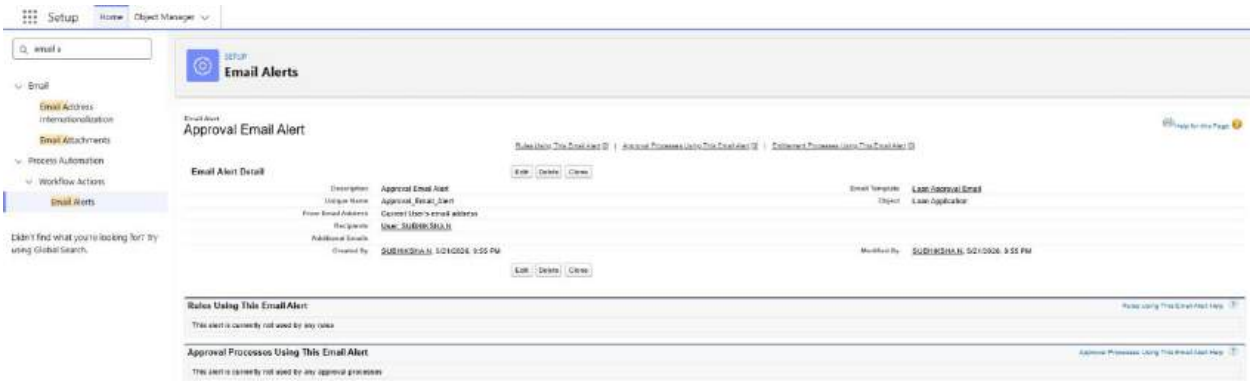


Figure 3.3.3: Approval Email Alert

Figure 3.3.3 illustrates the Approval Email Alert configured in Salesforce. The alert automatically sends the approval email to the customer when the loan application status is updated to Approved, ensuring timely communication and improving the efficiency of the approval workflow.

3.3.3 Rejection Email Alert

The **Rejection Email Alert** is configured to automatically send the rejection email whenever the application status changes to **Rejected**. The automation ensures that customers receive immediate notification regarding the outcome of their applications while reducing manual communication efforts.

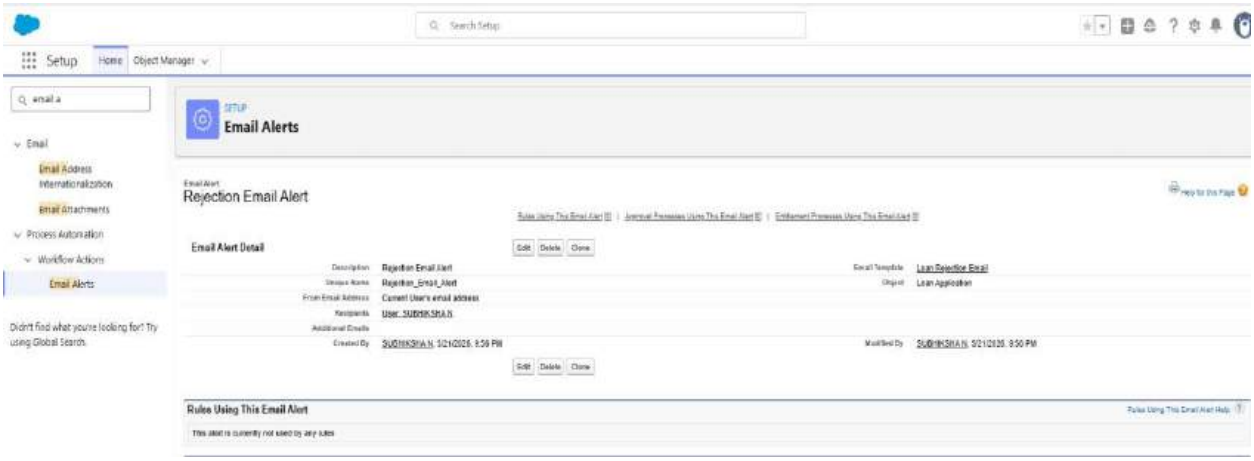


Figure 3.3.3: Rejection Email Alert

Figure 3.3.3 illustrates the Rejection Email Alert configured in Salesforce. The alert automatically sends the rejection email to the customer when the loan application status changes to Rejected, ensuring prompt communication, improved transparency, and a consistent notification process throughout the Automated Approval Workflow System.

4 IMPLEMENTATION – LOAN APPLICATION CRM & APPROVAL AUTOMATION

4.1 APPROVAL AUTOMATION

The **Approval Automation** module enables authorized loan officers and managers to review loan applications after submission. During the review process, the approver verifies the applicant's information, assesses loan eligibility, examines the supporting documents, and records comments based on the evaluation. Depending on the review outcome, the application may be **approved**, **rejected**, or **returned to the applicant** for further modifications.

Once the approval decision is completed, Salesforce automatically updates the application status, records the approval history, and triggers the appropriate workflow actions. If the application is approved or rejected, the system generates the corresponding notification and automatically sends an email to the customer, ensuring timely communication and improved transparency throughout the loan approval process.

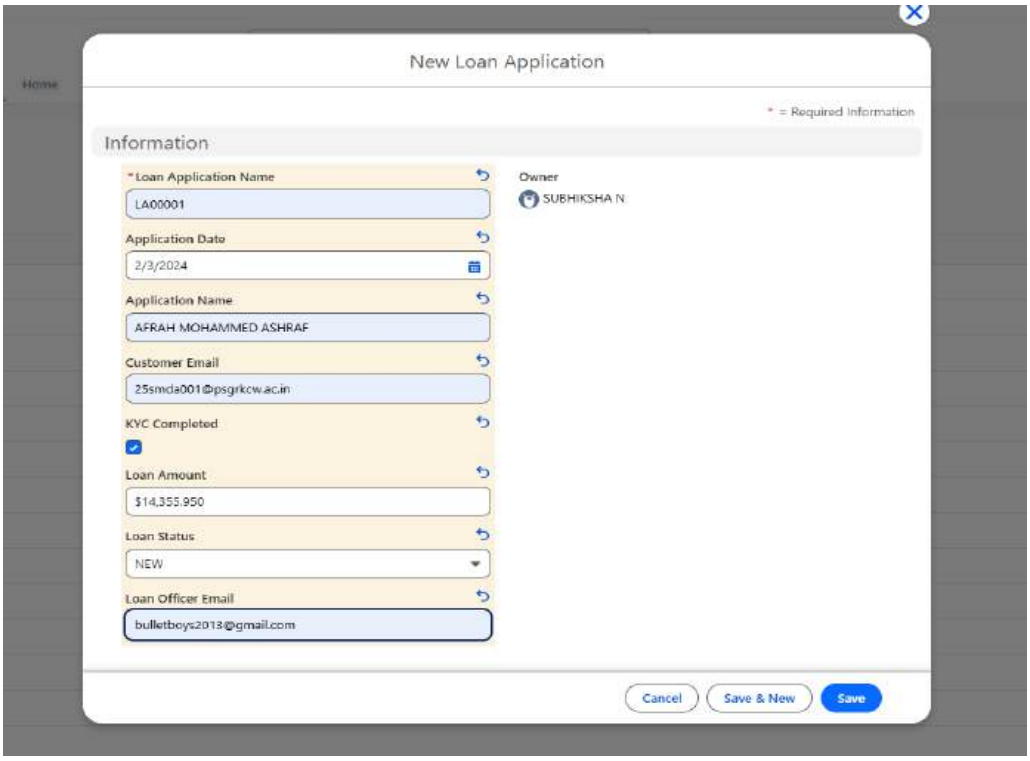


Figure 4.1.1: Loan Application Registration

Figure 4.1.1 illustrates the loan application registration screen of the Automated Approval Workflow System. This screen is used to register a new loan application by entering applicant details such as name, application date, loan amount, occupation, contact information, and other required information. After all mandatory fields are completed, the application is saved for further processing.

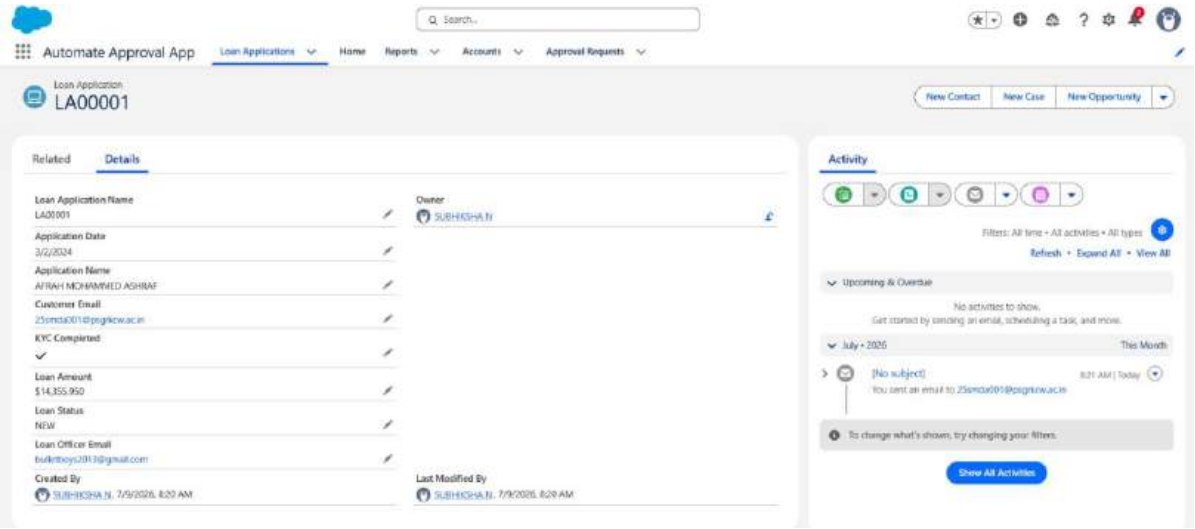


Figure 4.1.2: Loan Application Details

Figure 4.1.2 displays the complete details of the registered loan application. Users can review or update applicant information, verify the entered data, and monitor the current status of the application before submitting it for approval.

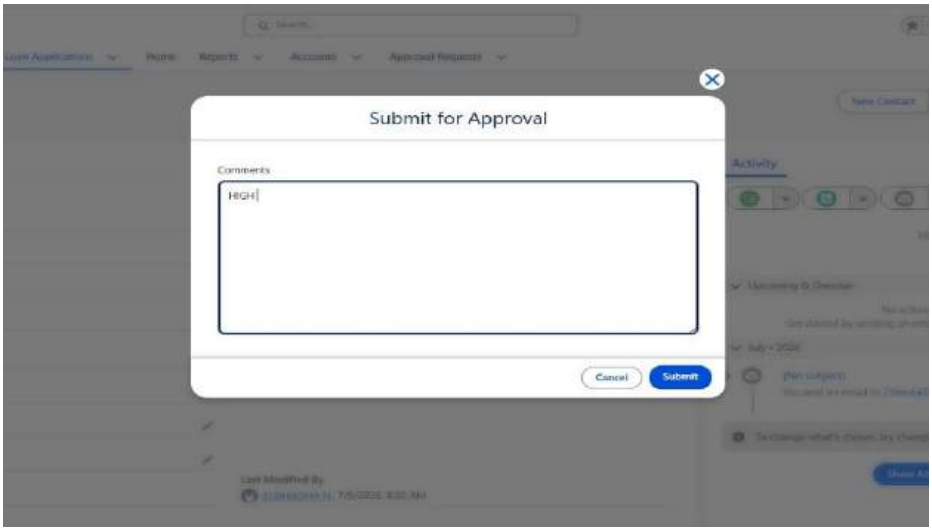


Figure 4.1.3: Loan Application Submission for Approval

Figure 4.1.3 illustrates the loan application submission process. After verifying the application details, the user submits the application for approval by entering review comments and selecting the approval priority, such as Low, Medium, or High. The application is then automatically forwarded to the designated approver through the Salesforce Approval Process.

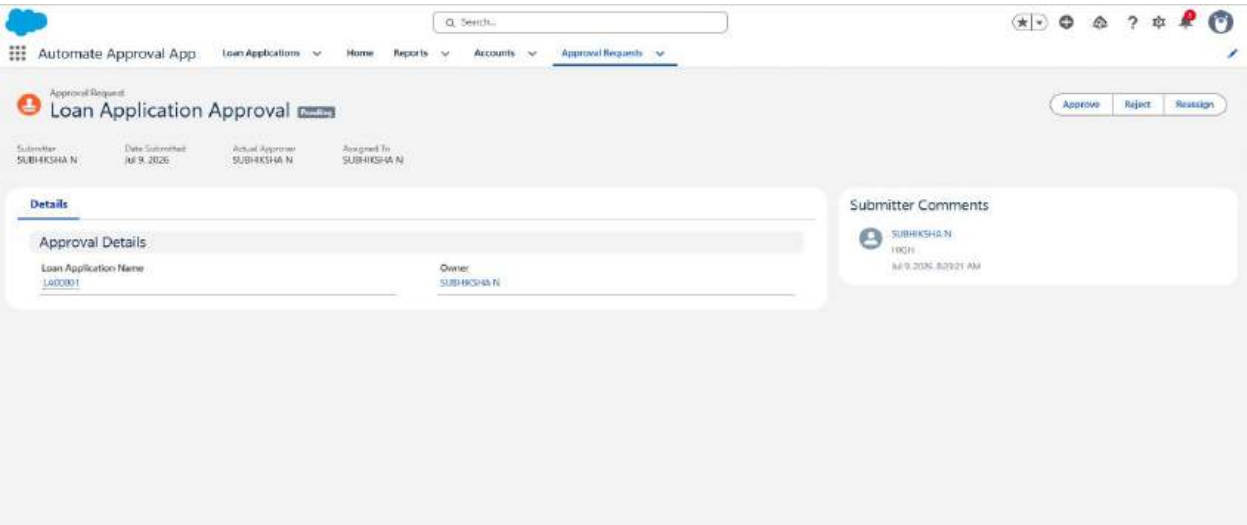


Figure 4.1.4: Pending Approval Status

Figure 4.1.4 shows that the loan application has been successfully submitted and is awaiting the approver's decision. The screen displays the current approval status, submission details, and comments entered by the applicant or loan officer, allowing users to monitor the progress of the approval request.

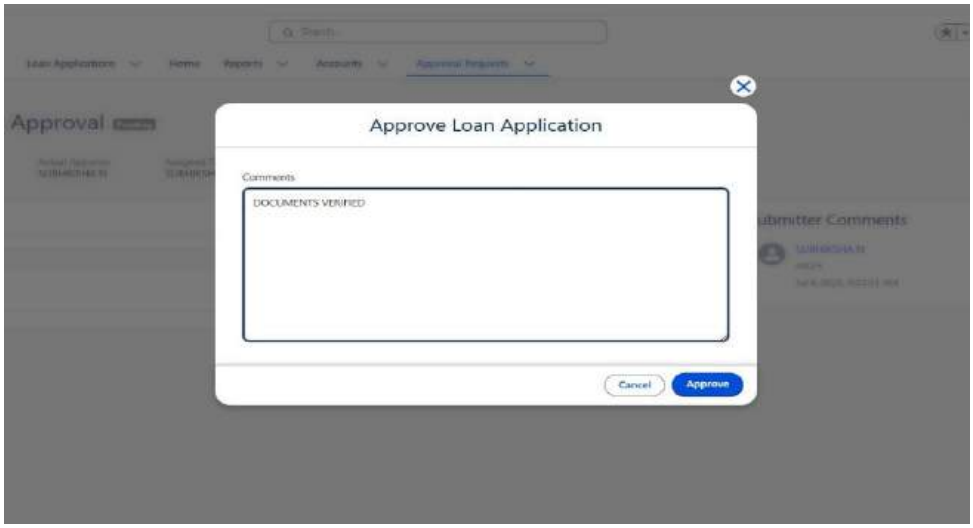


Figure 4.1.5: Loan Officer Notification

Figure 4.1.5 illustrates the notification received by the assigned loan officer after a loan application has been submitted. The loan officer reviews the application details, verifies the submitted information, and provides comments before proceeding with the approval decision.

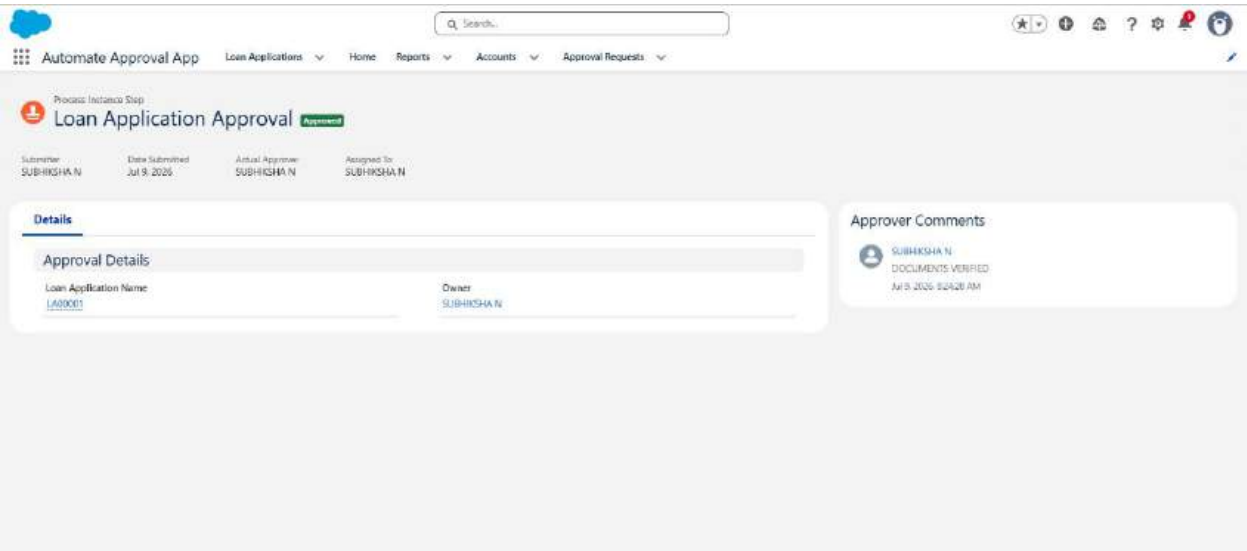


Figure 4.1.6: Loan Approval Decision

Figure 4.1.6 illustrates the approval decision screen used by the approver. The approver reviews the application details, enters approval comments, and approves the loan application based on the organization's approval policies. Once approved, Salesforce automatically updates the application status and records the approval history.

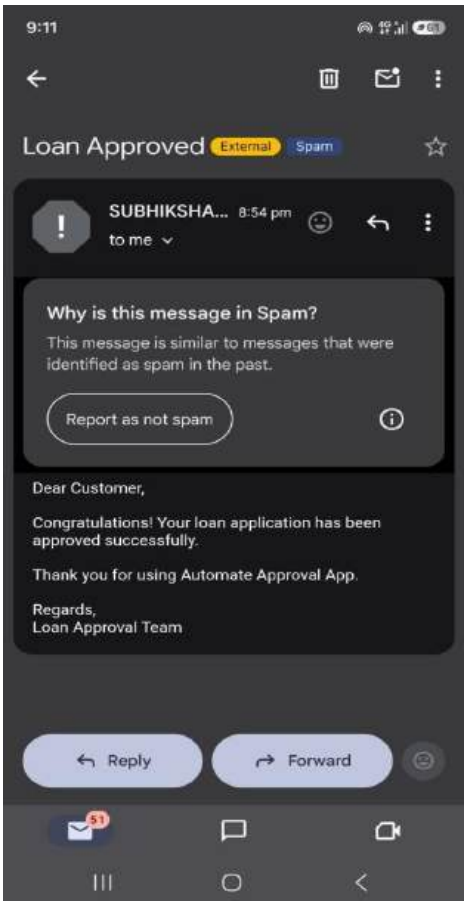


Figure 4.1.7: Approval Email Notification

Figure 4.1.7 illustrates the automated approval email notification generated by Salesforce after the loan application has been approved. The system automatically sends an email to the customer confirming the approval decision and providing relevant application details. This automated notification ensures timely communication, enhances transparency, and improves the overall customer experience.

4.2 REJECTION AUTOMATION

The **Rejection Automation** module is used when a loan application does not satisfy the organization's approval criteria. After reviewing the applicant's personal information, financial details, eligibility, and supporting documents, the authorized loan officer records the reason for rejection and updates the application status accordingly. Salesforce automatically stores the rejection comments, maintains the approval history, and updates the application record for future reference.

Once the rejection process is completed, the system automatically triggers an email notification to inform the customer about the rejection decision. This automated workflow ensures transparency, maintains proper documentation, reduces manual effort, and enables timely communication with the applicant throughout the loan approval lifecycle.

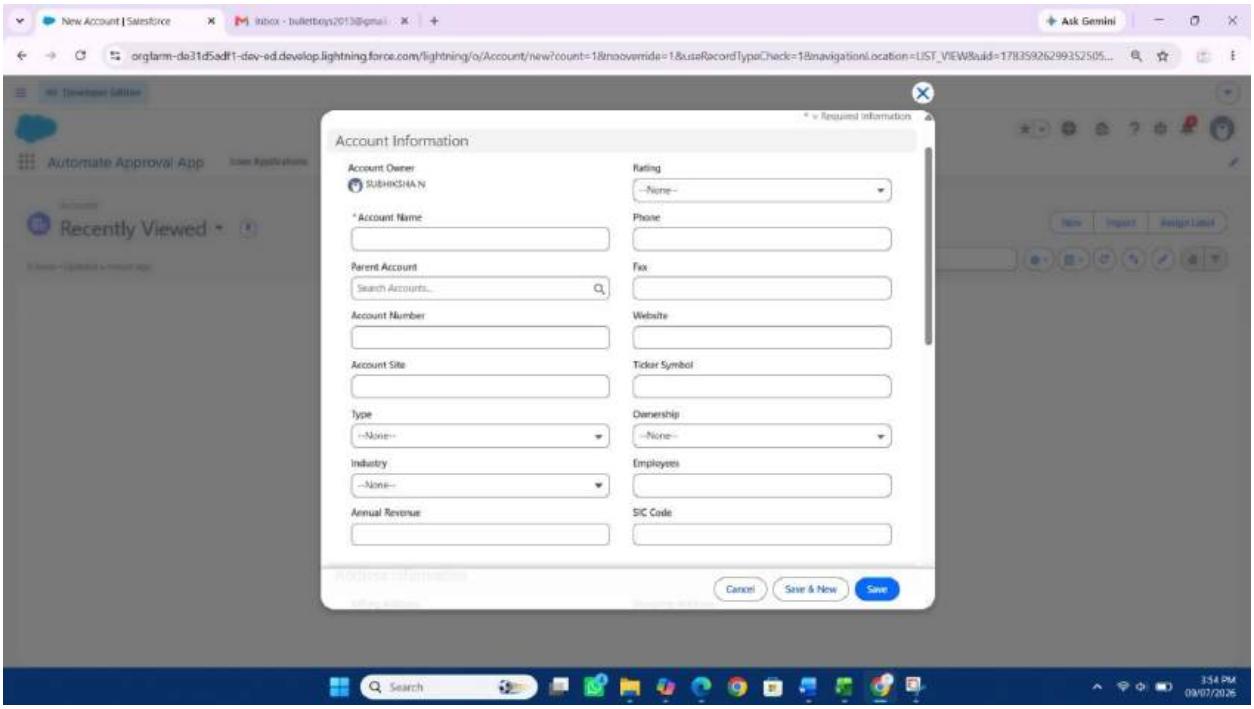


Figure 4.2.1: Loan Application Registration

Figure 4.2.1 illustrates the loan application registration screen used to create a new loan request. The applicant's personal, financial, and loan-related information is entered and securely stored in Salesforce for further processing and approval.

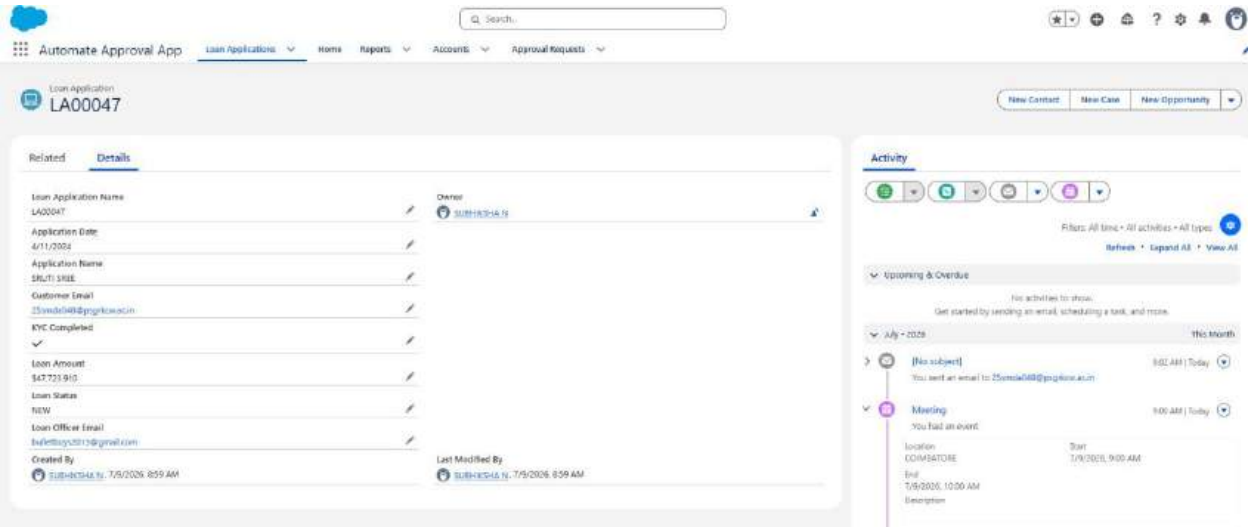


Figure 4.2.2: Loan Application Details

Figure 4.2.2 displays the complete details of the submitted loan application. Users can review the entered information, verify customer details, and ensure that all mandatory information is accurate before submitting the application for approval.

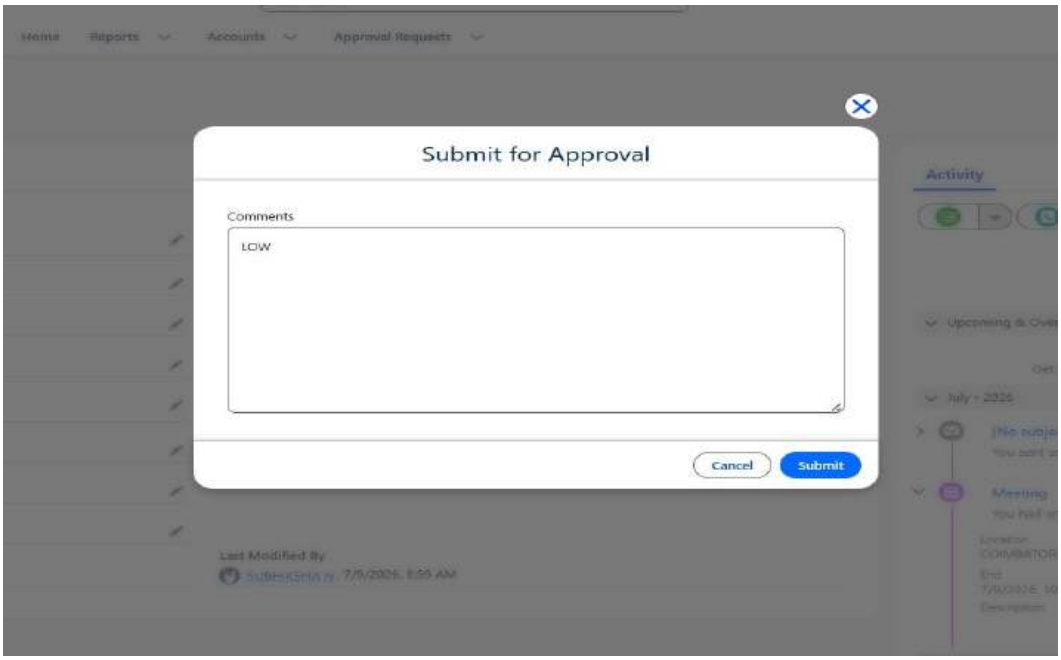


Figure 4.2.3: Loan Application Submission

Figure 4.2.3 illustrates the submission of the loan application for approval. After reviewing the application, the user selects the approval priority (Low, Medium, or High), enters the required comments, and submits the application to the assigned loan officer for review.

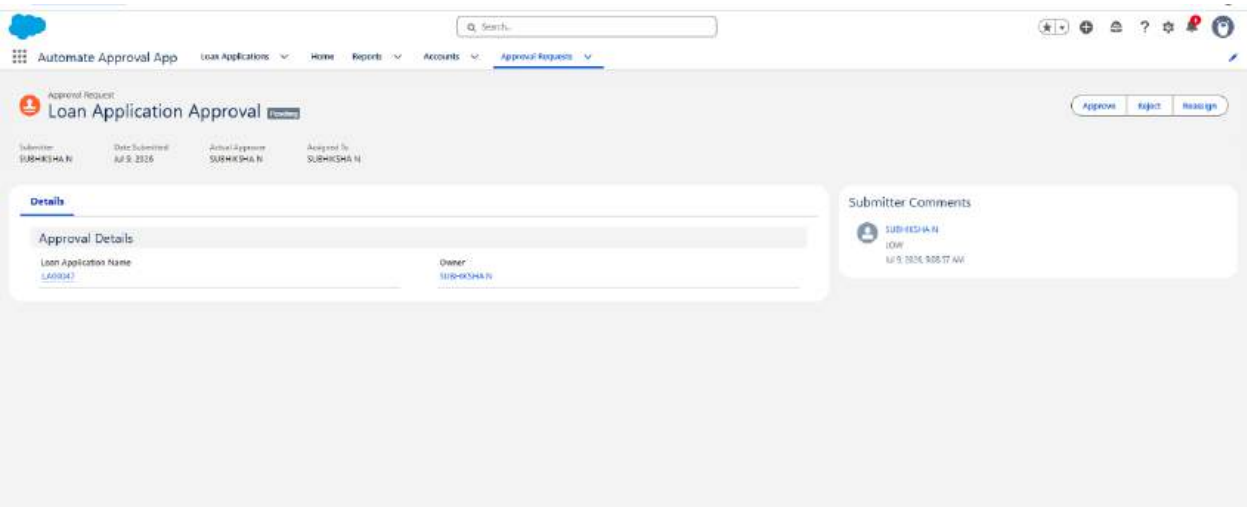


Figure 4.2.4: Pending Approval Status

Figure 4.2.4 shows that the loan application has been successfully submitted and is currently awaiting the loan officer's decision. The application remains in the Pending status until it is either approved or rejected by the authorized approver.

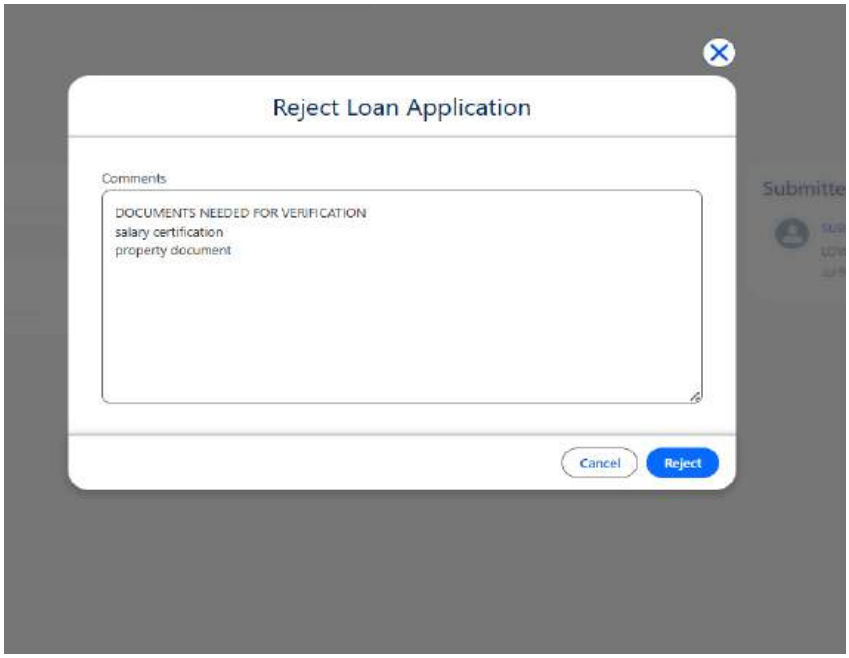


Figure 4.2.5: Rejection Review

Figure 4.2.5 illustrates the rejection review process, where the loan officer examines the application and records comments explaining the reasons for rejection. These comments are stored within Salesforce to maintain transparency and provide meaningful feedback for the applicant.

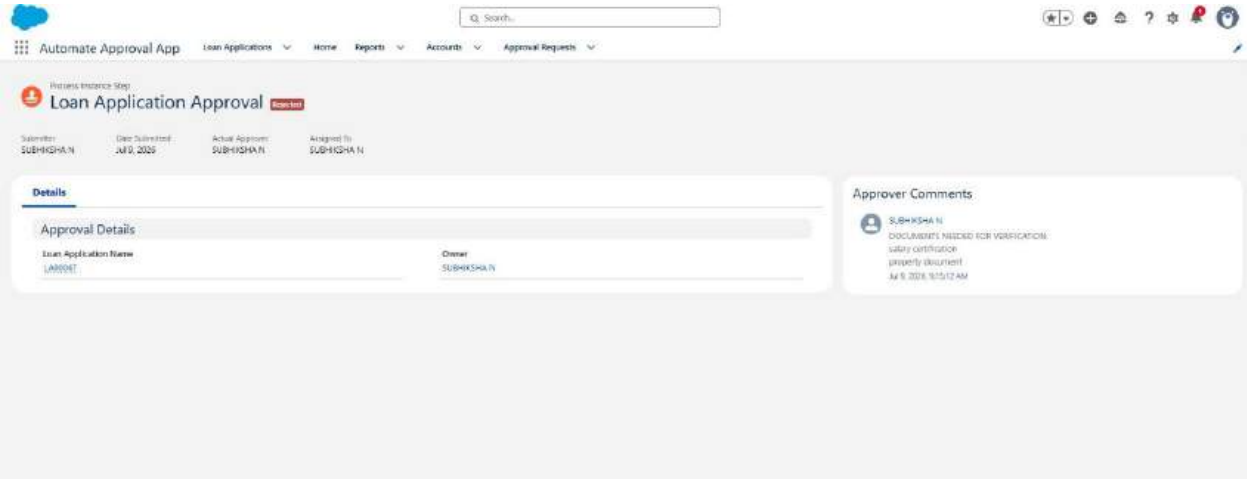


Figure 4.2.6: Loan Application Rejected

Figure 4.2.6 illustrates the rejected loan application. After completing the review process, the authorized loan officer rejects the application, and Salesforce automatically updates the application status while recording the rejection comments for future reference and auditing purposes.

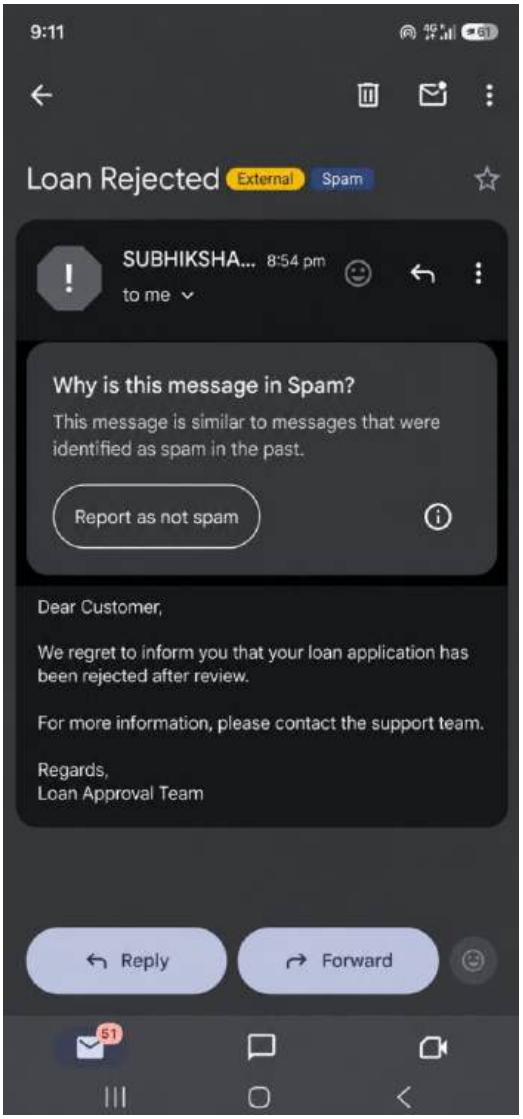


Figure 4.2.7: Rejection Email Notification

Figure 4.2.7 illustrates the automated rejection email notification generated by Salesforce. After the loan application is rejected, the system automatically sends an email to the customer informing them of the rejection decision, along with the relevant remarks or further instructions. This automation ensures prompt communication and enhances transparency throughout the approval process.

4.3 ACCOUNTS

The **Accounts** module is used to create and manage customer account information within the Automated Approval Workflow System. It enables users to register new customer accounts,

securely store account details, and maintain accurate customer records throughout the loan approval process.

Users can view account information, update customer records whenever necessary, and generate account-related reports for monitoring and administrative purposes. The module provides a centralized repository for customer information, ensuring that account data remains organized, accurate, and easily accessible.

By maintaining a structured customer database, the Accounts module supports efficient loan processing, improves data consistency, and enables seamless integration with loan applications, approval workflows, and reporting features within the Salesforce platform.

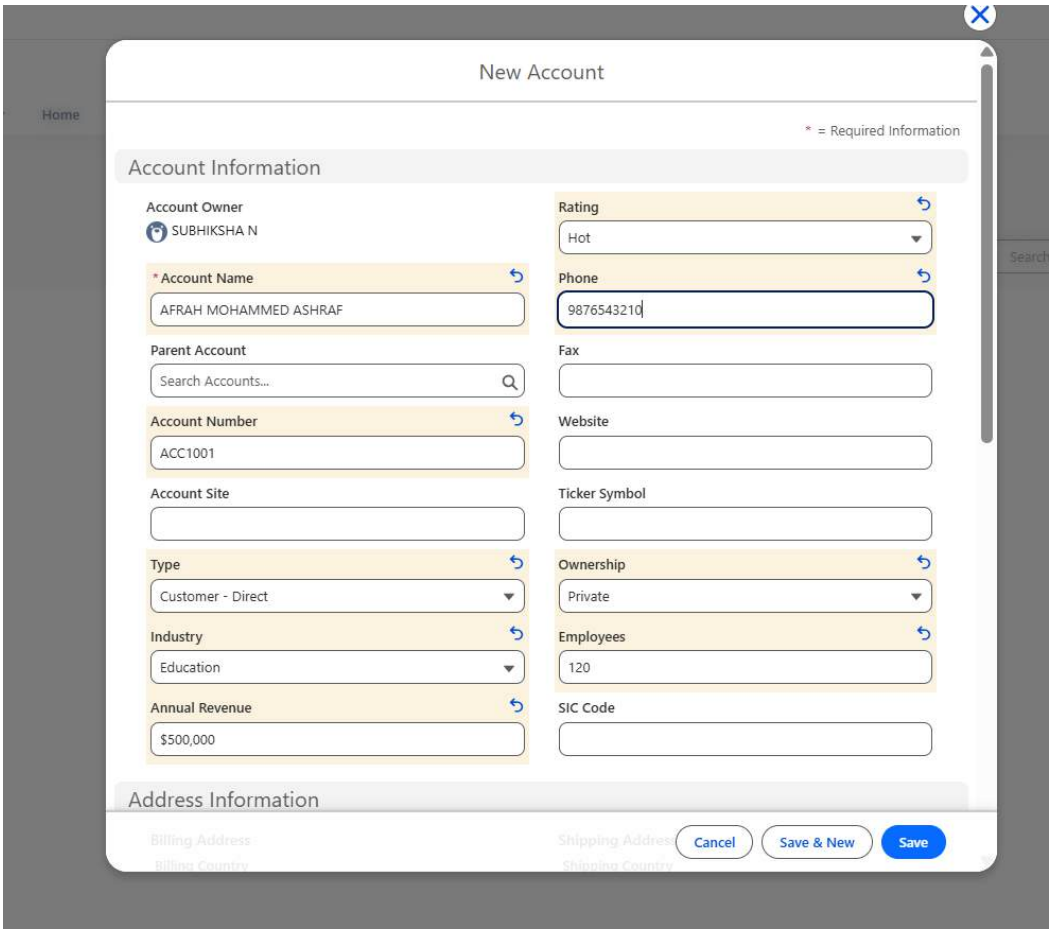


Figure 4.3.1: Customer Account Creation

Figure 4.3.1 illustrates the customer account creation screen used to register a new customer in the Automated Approval Workflow System. The user enters essential information such

as the customer name, account number, account type, contact information, and other required details. After the information is saved, a new customer account is successfully created and stored in the Salesforce platform.

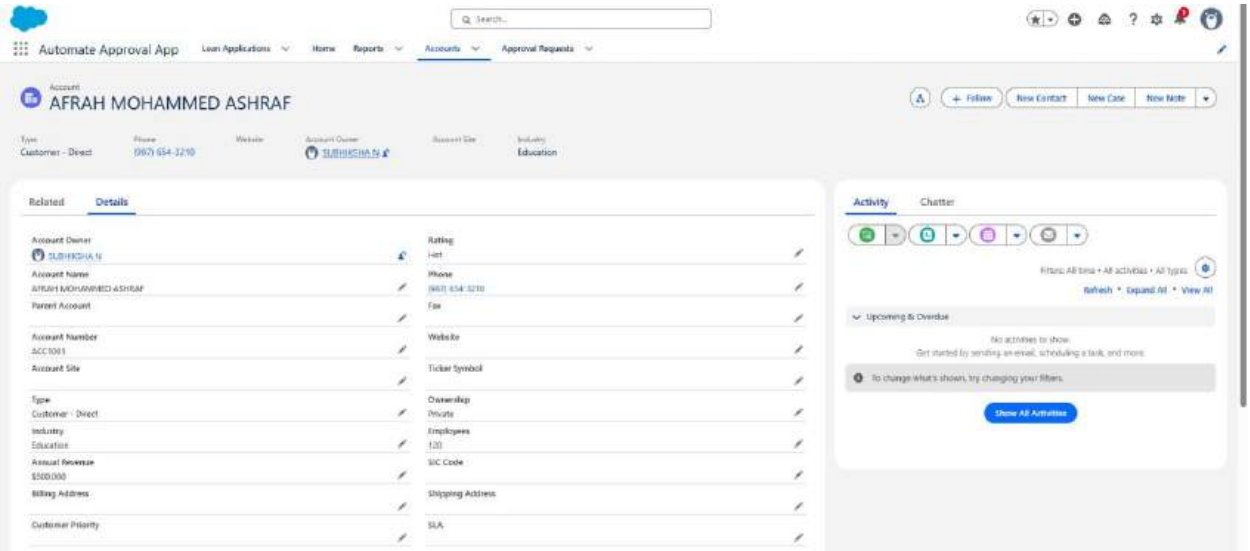


Figure 4.3.2: Customer Account Details

Figure 4.3.2 displays the complete details of the customer account. Users can review the stored account information, verify the entered data, and update customer details whenever necessary. This screen helps maintain accurate and up-to-date customer records within the system.

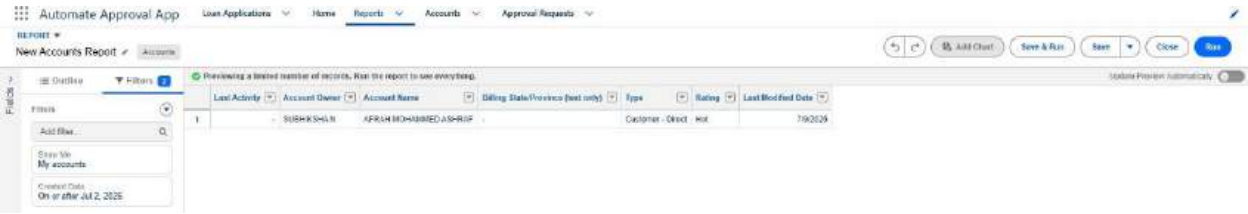


Figure 4.3.3: Customer Account Report

Figure 4.3.3 illustrates the Customer Account Report generated in Salesforce. The report displays all customer account records, including account information, customer details, and account status. It enables administrators to monitor customer records, maintain accurate account information, and manage customer-related activities efficiently.

4.3 REPORTS

4.3.1 Loan Approval Report

The **Loan Approval Report** displays all loan applications that have been approved through the automated approval workflow. The report includes information such as Loan Application ID, Customer Name, Loan Amount, Application Date, and Approval Status. It enables administrators and managers to monitor approved loans, analyze approval trends, and track successful loan processing activities within the system.

Dashboard Component

- Displays the total number of approved loan applications.
- Provides a summary of approved loan amounts.
- Helps monitor loan approval trends and workflow performance.

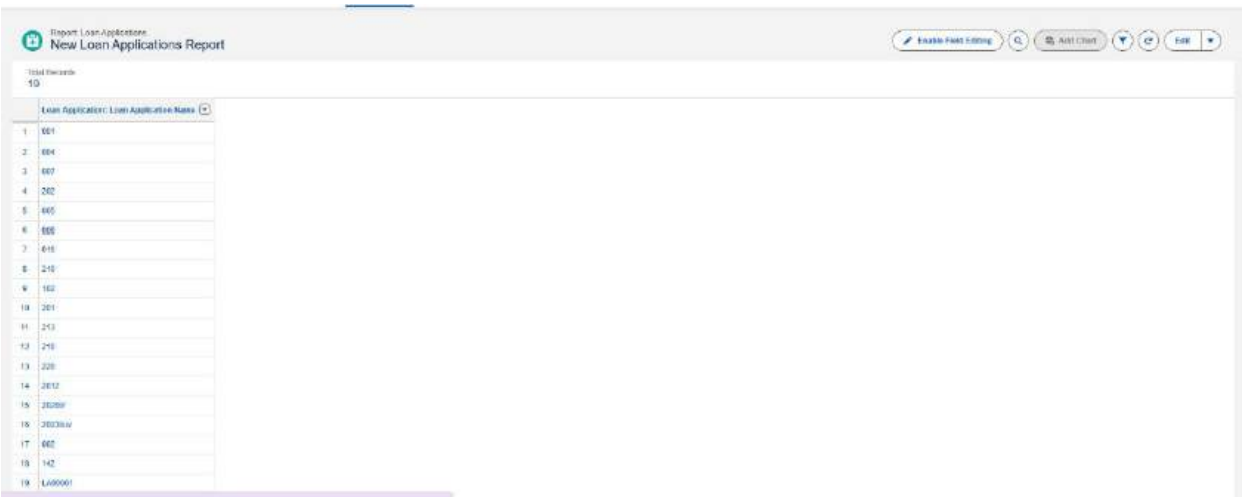


Figure 4.3.1: Loan Approval Report

Figure 4.3.1 displays the list of approved loan applications with customer details, loan information, approval status, and approval date. The report helps administrators monitor approved loans, generate reports, and evaluate the performance of the loan approval process.

4.3.2 Loan Rejection Report

The **Loan Rejection Report** contains details of loan applications that have been rejected during the approval process. The report includes customer information, loan amount, application date, and rejection status. It assists administrators in identifying rejection patterns, evaluating loan eligibility issues, and improving decision-making processes.

Dashboard Component

- Displays the total number of rejected loan applications.
- Provides visual analysis of rejection statistics.
- Helps identify trends and areas requiring process improvement.

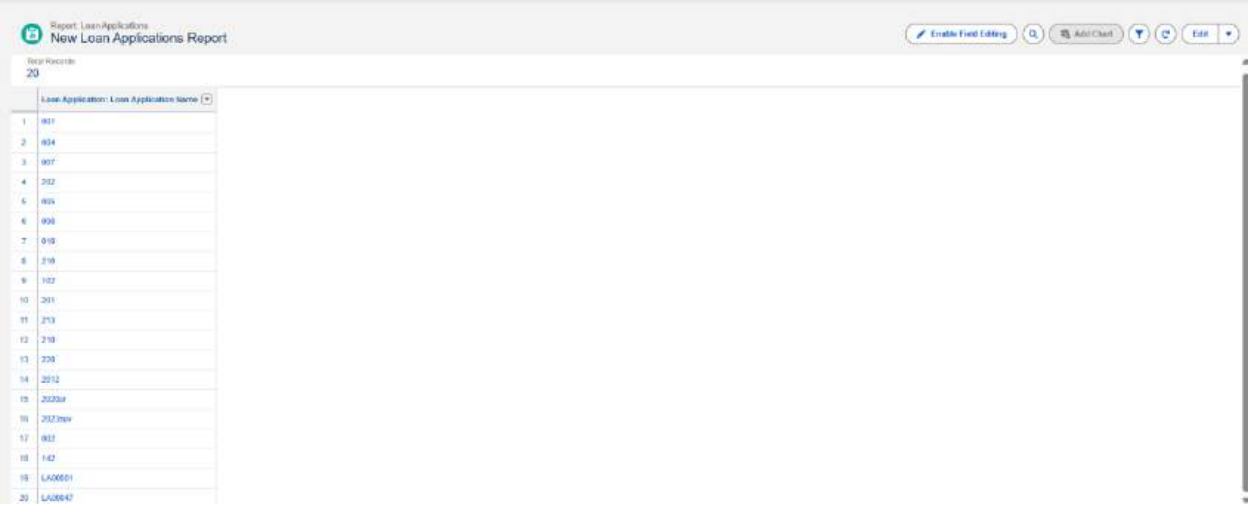


Figure 4.3.2: Loan Rejection Report

Figure 4.3.2 displays the list of rejected loan applications with customer details, loan information, rejection status, and rejection remarks. It helps administrators track rejected applications, analyze rejection patterns, and maintain complete records for future reference.

4.3.3 Customer Account Report

The **Customer Account Report** provides information about all customer accounts stored in the Salesforce system. The report includes Account Name, Account Number, Industry, Ownership, Annual Revenue, Employees, and Rating. It enables users to manage customer records, monitor account details, and maintain accurate customer information for loan processing activities.

Dashboard Component

- Displays the total number of customer accounts.
- Shows account distribution by industry and ownership.
- Provides insights into customer profiles and business categories.

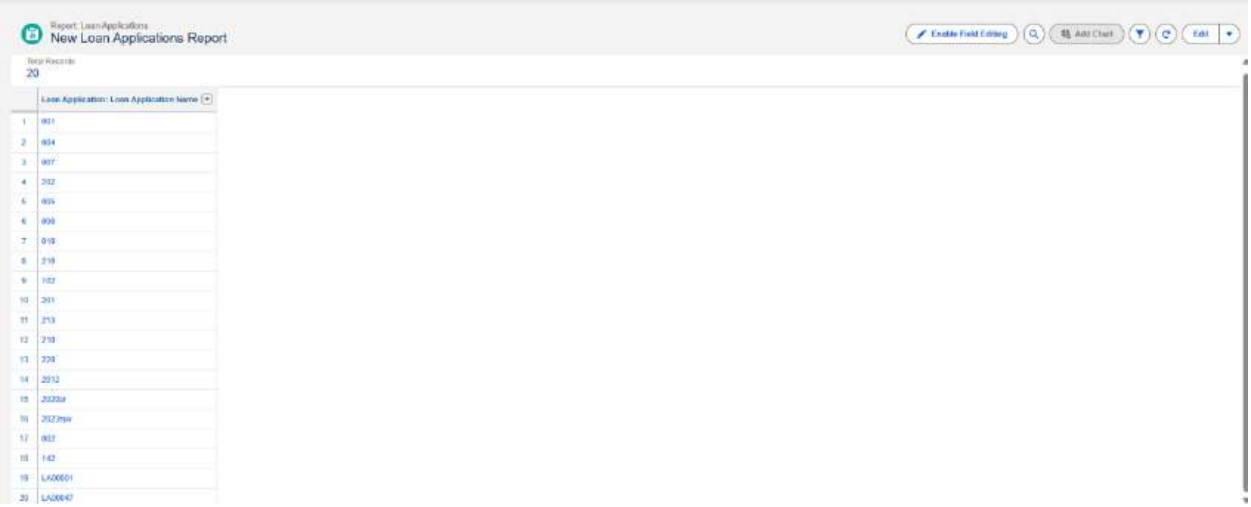


Figure 4.3.3: Customer Account Report

Figure 4.3.3 displays the Customer Account Report, listing all customer account records maintained in the Salesforce platform. The report enables users to view, track, and manage customer information efficiently while supporting loan processing, account management, and business reporting.

5 CONCLUSION

The Automated Approval Workflow System was successfully developed using the Salesforce CRM platform to automate the loan approval process. The application effectively replaces traditional manual approval procedures with an automated workflow, thereby improving the efficiency, accuracy, transparency, and reliability of loan processing activities. By utilizing Salesforce features such as Custom Objects, Approval Processes, Record-Triggered Flows, Email Templates, Email Alerts, Validation Rules, and Reports, the system streamlines the complete approval lifecycle while minimizing manual intervention.

The implemented solution enables customers to submit loan applications through a centralized platform, while loan officers and managers can efficiently review, verify, and process requests using automated approval workflows. The system automatically updates application status, records approval history, and sends email notifications to customers regarding the outcome of their applications. This automation significantly reduces processing time, improves communication, and enhances the overall customer experience.

The application also provides centralized management of customer accounts and loan application records, ensuring data consistency, security, and transparency throughout the approval process. Built-in validation rules, role-based security, and workflow automation improve data integrity while supporting organizational compliance and governance. Furthermore, the reporting capabilities enable administrators to monitor workflow performance and maintain complete visibility into approval activities.

Overall, the project demonstrates the practical implementation of Salesforce technologies in developing a secure, scalable, and efficient workflow automation solution. The Automated Approval Workflow System enhances operational productivity, reduces manual effort, improves decision-making, and supports organizations in delivering faster, more transparent, and customer-centric loan approval services.

6 FUTURE ENHANCEMENTS

The current implementation of the Automated Approval Workflow System provides an efficient and reliable solution for automating loan approval processes. However, the application can be further enhanced by incorporating additional technologies and advanced features to improve functionality, security, user experience, and overall business performance. Future enhancements include the following:

- Integration with external Know Your Customer (KYC) verification services.
- Integration with Credit Score APIs for automated eligibility assessment.
- SMS and WhatsApp notifications in addition to email alerts.
- AI-based loan eligibility prediction using customer data.
- Digital document verification and electronic signature (e-signature) support.
- Mobile application for customers, loan officers, and managers.
- Advanced real-time dashboards with interactive analytics and reporting.
- Integration with banking and payment systems for automated loan disbursement.
- Multi-level approval workflows for different categories of loans.
- Role-based dashboards for administrators, managers, and loan officers.

The implementation of these enhancements will further improve workflow automation, strengthen data security, support intelligent decision-making, and provide a more comprehensive, scalable, and user-friendly loan management solution.

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